

Estimation of Random Coefficient Demand Models: Two Empiricists' Perspective

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Abstract

We document the numerical challenges experienced when estimating random-coefficient Logit demand models introduced in the seminal work of BLP. We use two widely known datasets, various optimization algorithms, a large number of starting values, and different tolerances of the fixed-point iterations to document such challenges. We show that the optimization algorithms converge at points where the first- and second-order optimality conditions fail. We also provide cases of convergence at multiple local optima, as well as instances of convergence failure. Our findings indicate variation in the value of the objective function upon convergence that goes in hand with variation in parameter estimates and translates into variation in the models' economic predictions. Economic variables of interest, such as price elasticities, as well as changes in consumer welfare and firm profits following hypothetical merger exercises, vary widely for combinations of optimization algorithms and starting values that converged under tight tolerance for the fixed-point iterations. The minimum and maximum values of these economic variables of interest differ, at least, by a factor of 2 and up to a factor of 10.

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1 Introduction

Econometric modeling has become increasingly synonymous to the estimation of nonlinear models, where the objective function may not be globally concave or convex. This is very often the case in the so-called structural approach, which attempts to infer relationships between observable endogenous variables, and observable explanatory or unobservable variables based on optimizing behavior dictated by economic theory (Reiss and Wolak (2007)). Obtaining parameter estimates in these cases typically requires a nonlinear-search algorithm with a set of starting values and stopping rules for termination.

For the class of demand models introduced in the seminal work of Berry et al. (1995), henceforth, BLP, we find that convergence may occur in regions of the objective function where the first- and second-order optimality conditions fail. We also experience convergence at multiple local optima, as well as instances of convergence failure. Furthermore, parameter estimates and implied economic predictions, such as price elasticities, and changes in consumer welfare and firm profits due to hypothetical merger exercises, exhibit notable variation upon convergence of the nonlinear search.

The difficulties surrounding proofs for the existence of a global optimum of a criterion function for nonlinear extremum estimators are widely known and well-documented (McFadden and Newey (1994)). This is particularly true in the case of nonlinear GMM estimators like the one considered here. At the same time, the estimation problem that empirical economists, like ourselves, face very often involves technical difficulties with material implications for the conclusions of their analyses, which may not be publicized to the extent that they should. Based on our reading of the literature, this has been largely the case for the class of demand models we consider in this paper. The purpose of our work is to show that a thorough optimization exercise and a clear documentation of its design and challenges are potentially as important to the conclusions of an empirical exercise as the identification approach. The estimation of typical BLP random-coefficient (RC) Logit demand models based on two widely known data sets for cereals and automobiles gives us the opportunity to convey this message.

A typical BLP RC-Logit demand model achieves more flexible substitution patterns compared to the simple or nested Logit by allowing consumer heterogeneity in the valuation of product characteristics for differentiated products. At the same time, the model contains a product-specific demand shock in each market capturing all product characteristics that affect consumer choices but the econometrician cannot control for. Berry et al. (1995) introduced an estimation approach that also addresses endogeneity, as products with higher unmeasured quality most probably sell at a higher price, using GMM and fixed-point iter-

ations. These iterations allow the researcher to retrieve the product-specific demand shock in each market.

The BLP RC-Logit demand models, as well as their variants, are among the most popular state-of-the-art discrete-choice demand models and have provided answers to a variety of important questions in numerous empirical studies.¹ Measuring market power (Nevo (2001)), analyzing horizontal merger effects (Nevo (2000a)), evaluating international-trade policies (Berry et al. (1999)), welfare gains due to new products (Petrin (2002)), construction of price indices that account for quality changes and product introductions (Nevo (2003)), to only name a few, are among the many important economic questions that have been addressed using BLP-type demand models. Table 1 lists articles using the same class of models, which have been published in prominent general-interest economic journals, as well as in the leading industrial organization journal.

The articles cited in Table 1 reflect our attempt to include only those studies that utilized what we perceive as the main ingredients of the BLP approach in demand estimation for static models using aggregate data and a GMM framework to address endogeneity. From our point of view, these ingredients include heterogeneity in consumers’ valuation of product characteristics, as well as the unobserved product- and market-specific demand shock and the associated fixed-point iterations to retrieve it.² There are papers that have been published in the journals of Table 1, such as Goolsbee and Petrin (2004) and Bayer et al. (2007), which contain some but not all of the main ingredients of the BLP approach for demand estimation in a static framework. We don’t list these papers and, hence, we serve less than justice to a “broad” definition of BLP-type demand models.

Unfortunately, utilizing the desirable features of the BLP RC-Logit models requires the solution to a non-trivial optimization problem whose computational complexity has been publicized only recently. Starting with previous drafts of this paper, which were largely contemporaneous with the earlier drafts of Dube et al. (2011), and more recently with the papers of Judd and Skrainka (2011) and Skrainka (2011), the optimization challenges underlying the estimation of this class of models are by now well documented. More specifically, these challenges stem primarily from the combination of a nonlinear search to obtain parameters capturing heterogeneity in consumers’ valuation for product characteristics, and the fixed-point iterations to infer the product- and market-specific demand shock. This outer nonlinear search coupled with the fixed-point iterations has been, by far, the most popular approach in estimating BLP demand models to date.

¹As of December 26, 2011, Berry et al. (1995) had 517 cites according to the Social Sciences Citation Index (SSCI). On the same day, Nevo’s Practitioner’s Guide (Nevo (2000b)), which popularized the BLP-type demand models with its accompanying MATLAB code, had 96 cites using SSCI.

²We include articles that assume both a continuous and a finite distribution for consumer heterogeneity.

To highlight the difficulties in the solution of the optimization problem underlying the estimation of the BLP RC-Logit models, we use two widely known data sets, three classes of optimization algorithms with 50 different sets of starting values, and two different implementations of the fixed-point iterations. The first data set contains the pseudo-real data for cereals used by [Nevo \(2000b\)](#). The second data set, which is for automobiles, is the one used in [Berry et al. \(1995\)](#). We use eleven optimization algorithms that may be divided into three categories: derivative-based, deterministic direct-search, and stochastic direct-search. The two implementations of the fixed-point iterations differ in the tolerance that we use to declare convergence—we use a loose and a tight tolerance. Following the recommendations of [McCullough and Vinod \(2003\)](#), we discuss a number of diagnostics regarding first- and second-order optimality conditions at the terminal points of the nonlinear searches.

Our findings point to substantial variation in the value of the objective function of the underlying optimization problem both within and across algorithms for those combinations of starting values and fixed-point tolerances that converge. The variation is present in both datasets and slightly more pronounced in the case of loose tolerances. Although the derivative-based algorithms exhibit a superior performance relative to their direct-search counterparts in the case of cereals, this does not seem to be the case for automobiles. Two of the publicly available gradient-based optimization algorithms give rise to the smallest objective function value in both data sets. In addition, a non-negligible number of combinations of starting values, optimization algorithms, and fixed-point tolerances fail to converge; this is particularly true in the case of cereals with an almost equal split between loose and tight tolerances.

The variation in objective function values leads to substantial variation in parameter estimates, even when we limit our attention to the set of parameters that give rise to the smallest objective function value for each optimization algorithm under tight tolerance for the fixed-point iterations. Using the gradient norm, the Hessian eigenvalues, and a scale-invariant weighted-gradient stopping criterion, we identify a single local optimum and a saddle point in the case of cereals. Using the same criteria, we identify five local optima in the case of automobiles.

The variation in parameter estimates translates into variation in economic variables of interest for both datasets. We use own-price elasticities to document the variation in economic predictions at the product level. We use an aggregate elasticity, which we calculate simulating a price increase of all products, as well as the change in consumer welfare and firm profits following a hypothetical merger in the two industries to document variation in economic predictions at the market level. All these economic variables exhibit substantial variation even when we focus on those combinations of optimization algorithms and starting

values that converged under tight tolerance for the fixed-point iterations and remove any outliers.

In the case of cereals, the own-price elasticity of the product with the highest market share is between -2.47 and -1.34 among those combinations of optimization algorithms and starting values that converged, with an average of -1.98 and a standard deviation of 0.14. Similarly, the average aggregate elasticity across the 94 markets is between -1.78 and -0.84. The mean is -1.34 and the standard deviation is 0.11. The average change in profits across markets due to the hypothetical merger is between \$104.3m to \$229.7m with a mean of \$170.8m and a standard deviation of \$15.7m. The average change in consumer welfare across markets for the same exercise also exhibits substantial variation: -\$975m to -\$469m, with a mean of -\$671m and standard deviation of about \$60m.

For automobiles, the variation is more substantial. The range of the own-price elasticity for the product with the highest market share is -4.72 to -0.51 with a mean of -2.85 and standard deviation of 0.61 when we exclude a handful of observations exceeding zero. When we limit our attention to those combinations of algorithms, starting values, and fixed-point iterations under tight tolerance that gave rise to the five local optima, the own-price elasticities for the same product are between -3.41 and -1.86, excluding a single positive value associated with one of those optima. In addition, the average aggregate elasticity across 19 markets is between -1.73 and -0.52. The mean is -1.23 and the standard deviation is 0.21. If we focus on the aggregate elasticity values associated with the local optima, we see the values between -1.38 and -0.64.

The average change in profits due to the hypothetical merger for automobiles exhibits a range between \$302m and \$2,289m with a mean (standard deviation) of \$782m (\$343m). In the case of local optima, the average change in profits is as low as \$580m and as high as \$2,289m. The average change in consumer welfare for the same exercise is between -\$4,321m and -\$542m. The mean is -\$2,178 and the standard deviation is \$627m. Limiting our attention only to those values associated with the local optima, the range is between -\$4,307m and -\$1,829m. The values of the economic variables discussed here that are associated with the smallest local optimum appear to be outliers in the corresponding distributions that emerge from combinations of optimization algorithms, starting values, and fixed-point iterations.

The remainder of the paper is organized as follows. Section 2 provides an overview of the BLP RC-Logit demand model. Section 3 offers an outline of the methodology for the merger simulation and the calculation of the implied change in consumer welfare along with a discussion of some recent developments in issues surrounding this exercise. The details of the design of our optimization exercise are discussed in Section 4. Section 5 is an overview

of the data and the demand-model specifications we employ. Section 6 describes the optimization results documenting the variation in objective function values due to combinations of optimization algorithms, starting values, and tolerances for the fixed-point iterations. In Section 7, we illustrate the implication of such variation for economic variables of interest. We offer some conclusions and recommendations in the final section. The Appendix provides a discussion of the parameter estimates for the demand models we estimated and results for some additional optimization exercises we considered.

2 The Demand Model

In this section, we describe the standard BLP-type RC-Logit demand model with aggregate data. Following standard notation in the literature, we assume that a consumer i derives utility from a product j in market t that may be written as:

$$u_{ijt} = x_{jt}\beta_i - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt} = V_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where p_{jt} is the price of product j in market t , and x_{jt} is a (row) vector of non-price product characteristics. The vector ξ_{jt} captures the product-specific demand shock in each market. Each individual is assumed to choose one of the $1, \dots, J$ products available in the market, or to not purchase at all. The no-purchase option is usually termed the outside good and its associated utility is $u_{i0t} = \varepsilon_{i0t}$. The Logit error term ε_{ijt} is the first source of consumer heterogeneity in the utility function. The second source of consumer heterogeneity are the random coefficients α_i and β_i , which may be written as follows:

$$\begin{bmatrix} \alpha_i \\ \beta_i \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \Pi D_i + \Sigma v_i, \quad D_i \sim P_D(D), v_i \sim P_v(v). \quad (2)$$

The decomposition in (2) leads to terms that are common across consumers, such as α and β , as well as to terms D_i and v_i , which are vectors of observed and unobserved consumer characteristics that affect purchasing decisions and follow the distributions P_D and P_v , respectively. Although the presence of random coefficients is desirable due to their ability to generate more realistic substitution patterns, it has direct implications for the computational complexity of the model as we discuss below. After combining equations (1) and (2), we obtain:

$$\begin{aligned} u_{ijt} &= \delta_{jt}(x_{jt}, p_{jt}, \xi_{jt}; \theta_1) + \mu_{ijt}(x_{jt}, p_{jt}, D_i, v_i, \theta_2) + \varepsilon_{ijt} \\ \delta_{jt} &= x_{jt}\beta - \alpha p_{jt} + \xi_{jt}, \quad \mu_{ijt} = [p_{jt}, x_{jt}]'(\Pi D_i + \Sigma v_i). \end{aligned} \quad (3)$$

We use $[p_{jt}, x_{jt}]$ to denote a column vector of appropriate dimension. The mean utility associated with the consumption of good j that is common across consumers in market t is captured by δ_{jt} . Deviations from this mean utility are reflected in μ_{ijt} and ε_{ijt} . The vectors θ_1 and θ_2 differ in that the former contains α and β , while the latter contains the elements of matrices Π and Σ . Under independence of consumer idiosyncrasies for characteristics, the market share of product j is given by:

$$s_{jt}(x, p_t, \delta_t; \theta_2) = \int_{A_{jt}} dP(D, v, \varepsilon) = \int_{A_{jt}} dP_\varepsilon(\varepsilon) dP_v(v) dP_D(D) \quad (4)$$

$$A_{jt}(x, p_t, \delta_t; \theta_2) = \{(D_i, v_i, \varepsilon_{it}) | u_{ijt} \geq u_{ilt}\}, \forall l = 1, \dots, J.$$

In the share equation 4, x includes the characteristics of the products, while $p_t = (p_{1t}, \dots, p_{Jt})'$ and $\delta_t = (\delta_{1t}, \dots, \delta_{Jt})'$. The error term ε can be integrated out analytically in (4) giving rise to the well-known Logit probabilities. Given distributional assumptions for v and D , the integral associated with market shares is commonly evaluated using Monte Carlo integration assuming a number ns of individuals:

$$s_{jt}(x_{jt}, \delta_{jt}, \theta_2) = \frac{1}{ns} \sum_{i=1}^{ns} s_{ijt} = \frac{1}{ns} \sum_{i=1}^{ns} \frac{\exp(\delta_{jt} + \mu_{ijt})}{\sum_{j=1}^J \exp(\delta_{jt} + \mu_{ijt})} \quad (5)$$

The shock ξ_{jt} that was introduced by Berry (1994) plays the role of the structural error in a demand system. In its absence, the market shares given by (5) are smooth and continuous deterministic functions of the product characteristics and price. The presence of ξ implies likely endogeneity of prices because both consumers and firms observe ξ and therefore its value enters into the firms' pricing decisions. The standard approach in the literature to address endogeneity is nonlinear GMM with the identifying assumption:

$$E[\xi_{jt} | x_{jt}, z_{jt}] = 0, \quad (6)$$

where z_{jt} is an appropriate vector of excluded instruments. Given a vector of mean utilities δ , a sample analog of the moment condition can be constructed and the researcher may proceed with estimation. The vector of mean utilities δ is retrieved by equating the observed market shares from the data with those implied by the model for a given vector of parameters θ_2 :

$$s_{.t}^{obs} = s_{.t}^{pred}(x, p_t, \delta_t; \theta_2). \quad (7)$$

As opposed to the simple Logit and nested Logit, where analytical solutions for δ are available for the system of equations in (7), the RC-Logit requires a numerical solution of a highly

nonlinear system of equations whose dimension equals the number of products in the market under consideration. The econometrician may retrieve δ using the following fixed-point iterations:

$$\delta_{.t}^{(k+1)} = \delta_{.t}^{(k)} + \ln s_{.t}^{obs} - \ln s_{.t}^{pred}(x, p_t, \delta_{.t}^{(k)}, \theta_2), \quad (8)$$

where $\delta_{.t}^{(k)}$ denotes the k th iterate. For a given value of θ_2 , the fixed-point iterations in (8) can be initiated with the Logit solution $\delta_{.t}^{(0)} = \ln(s_{.t}) - \ln(s_{0t})$, where s_{0t} is the share of the outside good and continues until some norm of the difference between two consecutive iterates is smaller than some pre-specified tolerance. Once δ is retrieved, ξ can be inferred from the following equation:

$$\xi_{jt} = \delta_{jt} - x_{jt}\beta - ap_{jt}. \quad (9)$$

The elements of θ_1 , namely α and β , in the last equation are retrieved using linear instrumental variables (IVs). Having defined $\theta = (\theta_1, \theta_2)$, and with the aggregate demand shock playing the role of a structural error term that is a function of θ , the econometrician faces a nonlinear GMM problem with objective function given by:

$$Q_T(\theta) = \{T^{-1}\xi(\theta)'Z\}W_T\{T^{-1}Z'\xi(\theta)\} \quad (10)$$

for an appropriate weighting matrix W_T assuming a sample size T . Inference is performed using results in [Berry et al. \(1995\)](#), with the asymptotics working as $J \rightarrow \infty$; [Berry et al. \(2004\)](#) offer additional details. The methodology just described allows the econometrician to perform a nonlinear search in the parameter space only for θ_2 by concentrating out θ_1 . This is feasible because, for a given value of θ_2 , we infer δ using (7) and (8) and given δ we obtain θ_1 using linear IVs. Having δ and θ_1 available, the researcher constructs the econometric error that appears in (10). Draws from P_v and P_D required in (5) are made once and are kept constant through the estimation exercise.

The algorithm just described, which consists of an “outer” loop that minimizes the objective function with respect to θ_2 and an “inner” loop that uses fixed-point iterations to infer δ , may be termed a Nested Fixed Point (NFP) algorithm in the language of [Rust \(1987\)](#). We follow this nomenclature in the remainder of our discussion. The publication of computer code by [Nevo \(2000b\)](#) undoubtedly contributed to the popularity of the NFP algorithm for the estimation of BLP-type demand models for the last ten years or so. Recently, studies have identified issues regarding computational aspects of the methodology outlined here.

[Dube et al. \(2011\)](#) show that the temptation to implement loose stopping criteria for the fixed-point iterations to speed up the estimation process may cause two types of problems. First, the approximation error of the inner fixed-point iterations propagates into the outer

GMM objective function and its derivatives that may cause an optimization routine to fail to converge. Second, to induce the convergence of an optimization routine, the researcher may then loosen the outer-loop stopping criterion. Consequently, even when an optimization run converges, it may falsely stop at a point that is not a local minimum. The authors offer an alternative formulation of the GMM problem as a Mathematical Program with Equilibrium Constraints (MPEC) building on the results in [Su and Judd \(2011\)](#), who show that the MPEC and NFP algorithms produce estimators with similar statistical properties.

3 Merger Simulation and Consumer Welfare

In many empirical exercises, demand estimation serves as an immediate input to study the effects of changes in the structure of an industry, such as price increases implied by a merger, to name an example. At the same time, measures of changes in consumer welfare implied by the new market structure, such as compensating variation, are immediately available following the demand estimation exercise as our discussion below shows.

More specifically, with demand estimates available, the construction of a matrix of price derivatives emerging from the first-order conditions implied by profit maximization is straightforward. Combined with information on the ownership structure of the market and a model of competition, inferring marginal cost is possible. For example, under static Bertrand and constant marginal costs, the first-order conditions associated with the firms' profit-maximization problem imply:

$$p - mc = \Omega(p)^{-1} s(p), \quad (11)$$

where p is the price vector, $s(\cdot)$ is the vector of market shares and mc denotes the corresponding marginal costs. The dimension of these vectors is equal to the number of products available in the market, say J . The Ω matrix is the Hadamard product of the (transpose) of the matrix of the share derivatives with respect to prices, and an ownership structure matrix. The ownership structure matrix is of dimension $J \times J$ with its (i, j) element equal to 1 if products i and j are produced by the same firm and zero, otherwise. Because prices are observed and demand estimation allows us to retrieve the elements of Ω , estimates of marginal costs, $\widehat{mc}$, are directly obtained using (11).

A simple change of zeros and ones in the ownership structure matrix along with a series of additional assumptions allows the simulation of a change in the industry's structure, as the one implied by mergers among competitors (e.g., see [Nevo \(2001\)](#)). Simply put, a merger simulation implies the same Bertrand game with a smaller number of firms. The vector of

post-merger prices p^* is the solution to the following system of nonlinear equations:

$$p^* - \widehat{mc} = \hat{\Omega}^{post} (p^*)^{-1} \hat{s}(p^*). \quad (12)$$

The elements of $\hat{\Omega}^{post}$ reflect changes in the ownership structure implied by the hypothetical merger. Solving for the post-merger prices is equivalent to solving a system of nonlinear equations of dimension equal to the number of products in the market under consideration. For example, using the cereal data set, we have 94 markets with 24 products in each market. As a result, solving (12) requires the solution of 94 systems of nonlinear equations of dimension 24. An approximate solution for the post-merger prices, which avoids the need to solve the systems of nonlinear equations and is discussed in [Nevo \(1997\)](#) is given by:

$$p^{approx} = \widehat{mc} + \hat{\Omega}^{post} (p^{pre})^{-1} \hat{s}(p^{pre}), \quad (13)$$

where $\hat{s}(p^{pre})$ is the pre-merger vector of market shares, and the elements of $\hat{\Omega}$ associated with share price derivatives are evaluated at the pre-merger prices. Thus, we avoid dealing with issues related to potential numerical instabilities of Newton routines used in the solution of nonlinear first-order conditions, as well as with issues related to the existence and the uniqueness of equilibrium.

To the best of our knowledge, in the case of Bertrand competition with multi-product firms facing RC-Logit demand functions of the type discussed here, there is no result showing: (1) existence of an equilibrium in pure strategies, and (2) whether the equilibria are unique solutions to the system of equations implied by the first-order conditions (FOCs) of the underlying game. In the papers we are aware of, both existence and uniqueness have been assumed (e.g., footnote 12 in [Berry et al. \(1995\)](#)).³

With the post-merger prices in hand, we can estimate expected consumer welfare changes due to the mergers under consideration. One such measure of change in consumer welfare is the compensating variation. Assuming away nonlinear income effects, as is the case in the demand models we consider here, following [McFadden \(1981\)](#) and [Small and Rosen \(1985\)](#),

³[Allon et al. \(2011\)](#) provide a sufficient condition under which a Bertrand equilibrium exists and the set of Bertrand equilibria coincides with the solutions of FOCs in the case of *single-product* firms facing RC-Logit demand functions. This condition precludes a very high degree of market concentration: no firm captures more than 50% of the potential market in any of the consumer segments that it serves. A somewhat stronger version of the same condition, namely, firm shares below 30%, establishes *uniqueness*. Allon et al. also provide a sufficient condition for a (unique) equilibrium for markets with an arbitrary degree of concentration in the presence of an exogenous price limit. However, in this case, the equilibrium may not necessarily reside in the interior of the feasible price region and, hence, not be characterized by the FOCs.

the compensating variation for individual i is given by:

$$CV_i = \frac{\ln \left[\sum_{j=0}^{j=J} \exp(V_{ij}^{post}) \right] - \ln \left[\sum_{j=0}^{j=J} \exp(V_{ij}^{pre}) \right]}{\alpha_i}, \quad (14)$$

where V_{ij}^{pre} and V_{ij}^{post} are defined in (1) using the pre- and post-merger prices. Integrating over the density of consumers yields the average compensating variation in the population. Subsequent multiplication by the total number of consumers results in the total compensating variation for the population.

4 Optimization Design

We estimated the cereal and automobile demand models adapting the code used by Nevo (2000b), written in the MATLAB matrix language developed by Mathworks.⁴ The main body of Nevo’s code had to be altered to accommodate the setup of an exercise that involved 11 optimization algorithms using 50 sets of starting values and various stopping rules described below.⁵

The starting values for the mean utility vector δ are the fitted values of a simple Logit after adding draws from a zero-mean normal distribution with a standard deviation equal to the standard error of the Logit regression; therefore, the variation in the starting values represents regression error plausibly obtained across researchers. The starting values for the vector θ_2 entering the nonlinear part μ_{ijt} of the utility function in (3) are draws from a standard normal distribution; this represents the fact that little is known about the magnitude of θ_2 *a priori*.

Table 2 lists the 11 optimization algorithms we used to estimate the cereal and automobile demand models. The same table contains an acronym for each of the algorithms that we will use for the remainder of our discussion when we refer to them. These 11 algorithms are either derivative-based or direct-search. Five of the algorithms are derivative based. The remaining six are either deterministic or stochastic direct-search. The derivative-based algorithms utilize some information about the steepness and the curvature of the objective function, without necessarily keeping track of information associated with the Hessian matrix

⁴Nevo’s original code is available at <http://faculty.wcas.northwestern.edu/~ane686>. Our adaptation of Nevo’s original code is available at <http://web.mit.edu/knittel/www>.

⁵Among the papers listed in Table 1 and excluding Nevo’s writings, Copeland et al. (2011) [footnote 14], Nakamura and Zerom (2010) [footnote 26], Villas-Boas (2007) [page 631], Rekkas (2007) [footnote 5], and Villas-Boas (2007) [page 31] have also used Nevo’s MATLAB code, or modifications of it, to estimate BLP-type demand models. Chu (2010) [page 746], Villas-Boas (2007) [page 637], and Berry et al. (1999) [footnote 26] mention the use of multiple starting values for their nonlinear search algorithms explicitly.

while searching for a minimum of the objective function. The direct-search algorithms are based on function evaluations and are divided into deterministic and stochastic depending on whether or not they include a random component in their searches of the optimum of the objective function.

All of the algorithms are coded in MATLAB. Seven of the algorithms are commercially available as part of the MATLAB Optimization and Genetic Algorithm and Direct Search (GADS) toolboxes. The codes for the remaining four of the algorithms are publicly available from their authors. Two of the derivative-based algorithms (DER1-QN1 and DER2-QN2) are Quasi-Newton, the third (DER3-CGR) is a conjugate gradient, while the fourth (DER4-SOL) is an implementation of Shor’s r-algorithm. The last of the derivative-based algorithms (DER5-KNI) implements interior-point and active-set methods for solving continuous, non-linear optimization problems.

The MATLAB routines for the two Quasi-Newton algorithms, DER1-QN1 and DER2-QN2, are available in the MATLAB optimization toolbox and on the website maintained by Hans Bruun Nielsen, respectively. The routine for the conjugate-gradient algorithm, DER3-CGR, is also posted on Niensens’ website. Alexei Kuntsevich and Franz Kappel provide the routines for DER4-SOL.⁶ The KNITRO routines for DER5-KNI are available in the MATLAB optimization toolbox as add-ons. For the purpose of estimation, the derivative-based algorithms were implemented using analytical gradients and numerical Hessians (when necessary).

The routines for the three deterministic direct-search algorithms are available in the MATLAB optimization and Genetic Algorithm and Direct-Search (GADS) toolboxes, respectively. They include an application of the Nelder-Mead simplex, DIR1-SIM, the Mesh Adaptive Direct Search (MADS), DIR2-MAD, and the Generalized Pattern Search (GPS), DIR3-GPS. We refer the reader to [Lagarias et al. \(1998\)](#) for the mechanics of the Nelder-Mead simplex. [Torczon \(1997\)](#) provides a detailed description of the GPS. Material related to MADS, a generalization of the GPS algorithm, is available in [Audet and Dennis \(2006\)](#).

The direct-search routines implement two simulated-annealing and one genetic algorithms. The code for the first simulated-annealing algorithm, STO1-SIA, is our translation of the code originally developed for the GAUSS matrix language by E.G. Tsionas. The routines for the second simulated-annealing algorithm STO3-SIG, as well as for the genetic algorithm STO2-GAL, are available in the MATLAB GADS toolbox. We refer to [Dorsey and Mayer \(1995\)](#) and [Goffe et al. \(1994\)](#) for a compact discussion of the genetic and simulated

⁶[Burke et al. \(2007\)](#) provide a compact self-contained discussion of Shor’s r-algorithm. See [Kappel and Kuntsevich \(2000\)](#) for additional details. Recently, [Furlong \(2011\)](#) has reported a better performance for DER4-SOL relative to DER1-QN1 and DIR1-SIM (see below) in estimating a BLP-type demand model for hybrid vehicles.

annealing algorithms in the context of econometrics, respectively.⁷

We experimented with a number of stopping rules for the various optimization algorithms we employed. For the majority of the algorithms, but not all, convergence is dictated by the change in the objective function and the parameter vector (in some norm) between two consecutive iterations of an algorithm on the basis of a specified tolerance; the gradient norm is another metric. A maximum number of iterations or function evaluations are also employed as stopping rules. We used a tolerance of 1E-03 for changes in both the parameter vector and the objective function. We limited the number of function evaluations to 4,000. Imposing an upper bound on the number of function evaluations was largely dictated by the use of the direct-search algorithms, which tend to be more time-consuming relative to the gradient based algorithms. If an algorithm exceeded the maximum number of function evaluations it was terminated.⁸

A stopping rule is also required for the NFP iterations (see Equation 8), which introduce an additional layer of computational burden given their linear rate of convergence. More specifically, the NFP rate of convergence is measured by the Lipschitz constant, which is the norm of a matrix involving the own and cross demand elasticities with respect to the demand shock ξ (Dube et al. (2011)). We present results using a loose and a tight tolerance for the fixed-point iterations. The loose tolerance reflects the approach in Nevo (2000b). More specifically, the tolerance is initially set to 1E-06. After that, the tolerance level becomes less stringent by a factor of 10 every 50 iterations if $\|\theta_2^{k+1} - \theta_2^k\| \geq 0.01$, where k and $k + 1$ is used to denote two successive iterates of θ_2 . If $\|\theta_2^{k+1} - \theta_2^k\| < 0.01$ then the tolerance level is set to 1E-09 and no adjustment takes place. As a result, a loose (tight) tolerance is implemented when the parameter estimates are far from (close to) the solution. We also present results fixing the tolerance associated with the automobile and cereal data to 1E-16 and 1E-14, respectively, imposing an upper bound of 2,500 NFP iterations.⁹

⁷A review of the papers listed in Table 1 indicates that, when algorithms are discussed, they are of the classes we considered here. For example, Nevo (2001) [page 319] uses a Quasi-Newton method, which corresponds to the *fminunc* function of the MATLAB optimization toolbox, according to his code. Berry et al. (1995) [page 868] use the Nelder-Mead simplex algorithm. Petrin (2002) also uses the Nelder-Mead simplex algorithm according to the NBER working-paper version of his paper (see page 30). Villas-Boas (2007) [page 637] first uses a simplex search and then a gradient method. Armantier and Richard (2008) [page 902] use the FORTRAN IMSL library to estimate their model without specifying the function(s) they employ. According to the supplemental material on Econometrica's website, Goeree (2008) uses the double-precision FORTRAN IMSL functions *umpol* and *uminf*. The first function implements the Nelder-Mead simplex algorithm. The second function uses a Quasi-Newton algorithm with a finite-difference gradient.

⁸The Appendix provides results using a tolerance of 1E-06 for changes in the parameter vector and the objective function value for the derivative-based algorithms for both cereals and automobiles.

⁹Dube et al. recommend a best-practice tolerance of 1E-14 for the fixed point iterations. The script *invertshares* in the most recent version of their MATLAB code indicates an upper bound of 2,500 NFP iterations.

Finally, we simulated the market shares using 20 and 50 individual draws from a standard normal distribution for the cereal and automobile data, respectively. The number of draws in the case of cereals reflects the setting in Nevo’s code. For automobiles, the number of draws is representative of what we perceive to be a standard approach among practitioners.¹⁰ The standard-normal draws are made once in the beginning and are held fixed during estimation such that the limit theorems of [Pakes and Pollard \(1989\)](#) hold. Following [Nevo \(2000b\)](#), we use Monte Carlo integration to approximate the integrals associated with the market share calculations.

5 Data and Specifications

We use two datasets for implementing the BLP GMM algorithm. The first is the cereal pseudo-real dataset from [Nevo \(2000b\)](#). The second is the automobile dataset used by [Berry et al. \(1995\)](#) and [Berry et al. \(1999\)](#). Much of our motivation for the use of these data was due to the fact that they were publicly available at the time of our first draft.¹¹ The reader should also keep in mind that none of the exercises undertaken throughout the paper should be viewed as replication or validation exercises.

Starting with cereals, the data consist of 2,256 observations for 24 products (brands) in 47 cities over two quarters. The 24 brands are present in each of the 94 markets. Our specification of the demand equation is identical to Nevo’s. More precisely, the specification includes cereal brand dummies, which subsume product characteristics other than prices, as well as unobservable consumer characteristics interacted with a constant term, price, sugar content (sugar), and a mushy dummy indicating whether the cereal gets soggy in milk (mushy).

The specification also includes interactions of product characteristics with consumer demographics drawn from the Current Population Survey. For example, the price is interacted with the individual’s log of income (income), the log of income squared (income sq) and a child dummy indicating whether the individual is less than sixteen years old (child). The constant, sugar, and mushy are all interacted with income and age. The mean taste parameters associated with the constant, price, sugar, and mushy are retrieved using a minimum-distance

¹⁰For example, [Jiang et al. \(2009\)](#) [footnote 1] argue that the commonly used values in the literature are between 20 and 50.

¹¹To the best of our knowledge, the code for BLP 1995 is not publicly available. However, GAUSS code for BLP 1999 was publicly available at James Levinsohn’s website at the University of Michigan at the time of the first draft of this paper, circa Fall 2005. We used the GAUSS code to extract the data used in this paper. Table 1 in BLP 1995 and Table 2 in BLP 1999 contain descriptive statistics indicating that the authors used the same data for both papers.

procedure given the presence of the brand dummies.¹² The interaction of the product characteristics with the consumer unobservables and demographics give rise to 13 terms in total. Therefore, the outer loop of the BLP GMM algorithm involves a parameter vector θ_2 of dimension 13. The 24 coefficients associated with the brand dummies, corresponding to the elements of the vector θ_1 , are concentrated out and are retrieved using linear IVs.

The automobile data consist of 2,217 observations for all models marketed between 1971 and 1990 in the U.S. Each model/year combination is treated as a separate product and each year between 1971 and 1990 is treated as a separate market. The number of products in each market lies between 72 (market 4, 1974) and 150 (market 18, 1988). The 2,217 model/years represent 997 distinct models. We use 4 observable product characteristics other than price. The first is the ratio of the vehicle’s horsepower to its weight (HP/WT). The second is the vehicle space, which is measured as length times width (space). The third characteristic is a dummy indicating for whether the air-condition is standard or not (AC). The last characteristic is tens of miles per gallon of gasoline (MPG).¹³

Contrary to the cereal demand specification, the automobile demand specification is not identical to the in either BLP 1995 or in BLP 1999. In addition, both BLP 1995 and BLP 1999 estimate demand jointly with supply, but we don’t.¹⁴ In our specification, a constant term, the price and the four product characteristics of the previous paragraph enter the utility function. All of these variables, with the exception of space are also assigned random coefficients that correspond to the standard deviation of normal draws. Based on our specification, the outer loop of the BLP GMM algorithm for automobiles involves a parameter vector θ_2 of dimension 5. The 6 coefficients corresponding to the elements of the vector θ_1 , are concentrated out and are retrieved using linear IVs.

In terms of our identification strategy regarding cereals, we use the 44 instruments readily available in Nevo’s data set. In the case of automobiles, our identification strategy is similar to that in BLP 1995. Our instruments consist of the 5 non-price automobile characteristics, their sums across other automobiles produced by the same firm, as well as their sums across automobiles produced by the rival firms.

In terms of inference, the econometrician faces, in principle, three sources of error in the BLP demand models: sampling error in estimating market shares, simulation error in approximating the shares predicted by the model, and the underlying model error. There is

¹²See Section 4 in Nevo for additional information regarding the data and variables used in the demand specification. The results are available in Table I. See Section 3.5 for additional details about the minimum-distance procedure used to retrieve the mean taste parameters.

¹³For additional details regarding the data, see Section 7.1 in BLP 1995.

¹⁴The demand and supply specifications in the two papers are highly similar but not identical. See Table IV on page 876 in BLP 1995 and Table 5 on page 416 in BLP 1999 for comparisons.

no doubt that addressing properly all three sources of error is necessary to obtain the correct standard errors—see BLP 1995 and [Berry et al. \(2004\)](#) for a detailed discussion. However, inference does not fall within the scope of our paper. Although we report standard errors that account only for heteroskedasticity in the underlying model error for both the cereal and the automobile data, we do not discuss them.

6 Optimization Results

6.1 Objective Function Values

Figure 1 contains box-and-whisker (BaW) plots of the objective function values by optimization algorithm and NFP tolerance. The top panel refers to cereals and the bottom panel refers to automobiles.¹⁵ The naming convention for the algorithms on the vertical axis follows Table 2. Each point in a BaW plot is a combination of a starting value and an NFP tolerance for which the optimization algorithm under consideration converged on the basis of some stopping criteria. Depending on the algorithm’s implementation, convergence does not necessarily imply that the first- and second-order conditions for a local minimum are met (see our discussion below). The prominent vertical lines on the left part of the figures indicate the smallest objective function values across all such combinations.

The various algorithms implemented here use a number of similar but not identical stopping criteria. These criteria include, for example, the change in the parameter vector or the associated gradient, the change in the objective function value, maximum number of iterations or function evaluations. Based on such stopping criteria, the algorithms generate “exit” codes to indicate the conditions under which they stopped. The algorithms in the MATLAB optimization and GADS toolboxes produce a zero exit code to indicate that the maximum number of iterations or function evaluations was reached and a negative exit code to indicate their failure to converge. We utilize such exit codes to declare convergence. In the case of the deterministic and stochastic direct-search algorithms in these two toolboxes, we used exit codes exceeding zero to declare convergence.¹⁶ Once again, convergence is not necessarily synonymous to termination at a point where the first- and second-order conditions

¹⁵For each of the BaW plots, the boxes cover the interquartile range, from the lower quartile to the upper quartile, and contain a vertical white line indicating the median. The whiskers, denoted by horizontal lines, intend to cover most or all the range of the data. The left whisker extends to a value that is the lower quartile minus 1.5 times the interquartile range, or the minimum should this be smaller. The right whisker extends to a value that is the upper quartile plus 1.5 times the interquartile range, or at the maximum, if this is smaller. Data points outside the whiskers are represented with dots.

¹⁶The reader should refer to our publicly available code for a detailed treatment of exit codes of the various algorithms in order to declare convergence.

of a local minimum are met.

In the case of cereals, 883 combinations of starting values, optimization algorithms, and NFP tolerances led to convergence. Almost half of them (440) were associated with tight NFP tolerance. For the automobiles, 988 out of 1100 combinations converged with more than half of them (514) associated with tight NFP tolerance. All 100 combinations for the STO1-SIA algorithm did not converge for the cereal data set, while 51 of them failed to converge in the case of the automobiles. In all 51 cases, STO1-SIA reached the maximum number of function evaluations—23 times under loose NFP tolerance and 28 times under tight NFP tolerance. Furthermore, DIR1-SIM did not converge 69 times in the case of cereals— 33 times under loose NFP tolerance and 36 times under tight NFP tolerance. DIR1-SIM did not converge 49 times in the case of automobiles— 48 times under loose NFP tolerance and 1 time under tight NFP tolerance. In all these instances, DIR1-SIM reached the maximum number of function evaluations. In addition, DIR2-MAD did not converge in 28 cases for the cereal dataset. The algorithm reached the maximum number of function evaluations in all 28 cases with an equal split between loose and tight NFP tolerances.¹⁷

The BaW plots in Figure 1 account for extreme objective function values due to rather meaningless stopping points of various combinations of optimization algorithm and starting values. They also account for combinations of algorithms and starting values that gave rise to extraneous observations for market shares (e.g., NaNs).¹⁸ For example, in the case of cereals, the maximum across the 883 observations exceeds 89,000, the 75th percentile is 882.81, while the median is just 101.09. The minimum value is 4.56, which is consistent with the value reported by Dube et al. (2011). As a consequence, we constructed the cereal BaW plots using values that lie below 110, a value that is very close to the median under tight tolerance, which is equal to 105.53. In the case of automobiles, the maximum across the 988 observations lies above 62,000, the median is 226.94 and the 90th percentile is 1,555.98. The minimum value is 178.06. Hence, we constructed that BaW plots excluding values above 430, which is very close to the 90th percentile of the values under tight tolerance that is equal to 428.57. By truncating the distributions, we are obviously understating the variation in the values of the objective function.

¹⁷In the case of automobiles, DER5-KNI generated the exit codes -100 (-101) 34 (12) times. According to Appendix A in Waltz and Plantenga (2009), exit codes -100 to -199 indicate that “A feasible approximate solution was found.” More specifically, the exit code -100 implies that “No more progress can be made, but the stopping tests are closed to be satisfied (within a factor of 100) and so the current approximate solution is believed to be optimal.” The exit code -101 implies that “It is possible the approximate feasible solution is optimal, but perhaps the stopping tests cannot be satisfied because of degeneracy, ill-conditioning or bad scaling.”

¹⁸In the case of automobiles, the 50th set of starting values for STO3-SIG under both loose and tight NFP tolerance gave rise to such extraneous market shares. There were no similar instances of extraneous market shares for cereals.

In the case of cereals, we see substantial variation in the objective function values both across and within optimization algorithms. Overall, the derivative-based algorithms, which use analytical gradients, exhibit superior performance, in terms of reaching regions of the objective function with low values, relative to their deterministic or stochastic direct-search counterparts. Among the 464 values not exceeding 110, 238 of them are related to combinations with loose NFP tolerance. Only 6 (2) combinations of starting values and NFP tolerance for DIR3-GPS (STO3-SIG) do not exceed 110. Six combinations of optimization algorithms, starting values and NFP tolerances achieve the minimum value of 4.56. All these six combinations are associated with the derivative-based algorithms DER2-QN2, DER4-SOL, and DER5-KNI. Three of the six combinations use tight NFP tolerance. Interestingly enough, all 100 combinations of starting values and NFP tolerances for DER4-SOL give rise to the value of 4.56.

For DER1-QN1, the objective function values are between 19.55 and 105.82. The last of the derivative-based algorithms, DER3-CGR, exhibits values between 19.62 and 32.15. The deterministic direct-search algorithms achieve objective function values between 17.22 and 90.78. The values for DIR1-SIM range from 17.22 to 66.46, while those for DIR3-GPS are between 50.99 and 90.78. The objective function values for the stochastic-search algorithms also exhibit substantial variation—between 31.59 and 109.16. This range, which is similar to that of DER1-QN1 is largely attributable to 56 of these values associated with STO2-GAL. The two values associated with STO3-SIG are equal to 108.13.

The pattern of substantial variation in the objective function values across and within optimization algorithms is also present in the automobile data. Contrary to the cereal data, however, the derivative-based algorithms do not exhibit a superior performance relative to their deterministic or stochastic direct-search counterparts in terms of reaching regions of the parameter space with low objective function values once we focus on values not exceeding 430. There are 863 observations with values less than 430, with 463 of them associated with the tight NFP tolerance.

Four combinations of optimization algorithms and starting values under tight NFP tolerance led to the smallest objective function value of 178.06. Two of the algorithms are derivative-based, DER2-QN2 and DER4-SOL. The other two algorithms are deterministic direct-search, DIR1-SIM and DIR3-GPS. Recall that two of these algorithms, DER2-QN2 and DER4-SOL, also led to the minimum objective function value for the cereal data set. Additionally, DIR1-SIM seems to be able to identify the point with the smallest objective function value, which was not the case for the cereal demand model.

An interesting observation is that none of the 100 values for DER5-KNI falls below 277.62. The values for the derivative-based algorithms all lie below 338.49. Two pairs of algorithms

exhibit very similar behavior in terms of the range of the objective function values. In the first pair, DER1-QN1 and DER3-CGR, the values are between 215 and 338.49. In the second pair, DER2-QN2 and DER4-SOL, we see values from 178.06 to 257.47. In the case of the direct-search algorithms, the values for DIR1-SIM fall between 178.06 and 338.45, while those for DIR2-MAD lie between 215.05 and 428.57. The values for DIR3-GPS exhibit more variation; they are as high as 401.05. Moving to the stochastic-search algorithms, the values for STO2-GAL exhibit the least variation, 205.43 to 293.16. The other two stochastic algorithms, STO1-SIA and STO3-SIG give rise to values between 186.23 and 406.31.

Overall, the box-and-whisker plots for the objective function values in Figure 1 show variation both within algorithm and across algorithms. This variation, although ameliorated to some degree, continues to be present even with tight NFP tolerance. It is difficult to judge the relative performance of classes of algorithms since converging to points in the objective function that have low values, but are not the consistent root, does not necessarily yield results that are “closer” to the truth. Given the variation that continues to exist even under tight NFP tolerances, we would recommend practitioners to experiment not only with multiple starting values, but also with more than one class of algorithms when estimating BLP-type demand models and report their experiences.

6.2 Gradients and Hessians

In this section, we investigate whether the objective function values reported in Tables A.2 and A.4 of the Appendix correspond to *local* minima as opposed to other critical points, such as saddles, by examining the gradient (g) and the Hessian (H) of the objective function. We focus on results associated with tight NFP tolerance given the findings in Dube et al. (2011). As in the previous section, our discussion excludes DIR2-MAD and STO1-SIA, because these algorithms stopped by exceeding the maximum number of function evaluations with tight NFP tolerance. When discussing our results, the reader should keep in mind that finding the *global* optimum of a function, or even proving that a given local optimum is a global optimum, is a difficult problem.¹⁹ Furthermore, to the best of our knowledge, the discussion of gradient and Hessian diagnostics almost never appears in empirical work involving BLP-type demand models—for a notable recent exception, see Goldberg and Hellerstein (2011).²⁰

¹⁹There are three cases in which the problem is somewhat easier. The first is when the objective function has one critical point in its domain. The second is when the function is globally concave or convex in its domain (e.g., Simon and Blume (1994), page 55). The final case is the one in which the domain of the objective function is a compact subset, say C of $\mathbb{R}^n$, assuming an n th-dimensional parameter space. Based on Weierstrass’s Theorem, every continuous function whose domain is a compact subset C achieves its global maximum and its global minimum on C (e.g., Simon and Blume, page 823).

²⁰None of the papers listed in Table 1 provides any diagnostics regarding the gradient or the Hessian of the objective function when they discuss estimation results.

We measure the length of the analytical gradient using its inf-norm $\|g\|_\infty$. The inf-norm is equivalent to finding the maximum of the absolute values of the gradient elements. We refer to $\|g\|_\infty$ as the gradient norm for the remainder of our discussion. While Nevo’s code provides analytical expressions for the gradient, it does not provide analytical expressions for the Hessian of the objective function. Following what we perceive to be common practice, we evaluated H using the DER1-QN1 algorithm that offers *numerical approximations* to the Hessian as a by-product. In addition, we constructed a scale invariant weighted-gradient stopping criterion, $g'H^{-1}g$.²¹ Using the MATLAB built-in eigenvalue function (*eig*), we calculated the eigenvalues of H to determine whether it is positive definite.

For cereals, the implied gradient norm for the estimates in Table A.2 is between 0.009 (DER4-SOL) and 472.77 (DIR3-GPS). The Hessian is positive definite for all the algorithms that terminated successfully with the exception of STO2-GAL and STO3-SIG; see panel (a) in Table 3. The weighted-gradient criterion is almost identical to zero for DER2-QN2, DER4-SOL and DER5-KNI. Therefore, based on the first- and second-order optimality conditions, the objective function value of 4.56 implied by the parameter estimates in Table A.2 corresponds to a local minimum. Based on our discussion above, we may call the local minimum at 4.56 a “global” minimum although we cannot prove that we have found a global minimum. To the best of our knowledge, although tests for the null of whether a global optimum of a criterion function has been identified are available, they are not widely used by practitioners. This statement is particularly true for those practitioners undertaking empirical exercises involving BLP-type demand models.²²

We also examined the condition number of the Hessian at the global minimum because an ill-conditioned Hessian puts the accuracy of the reported results into question. More specifically, the Hessian condition number is given by $\kappa(H) = \lambda_{\max}(H)/\lambda_{\min}(H)$, where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and the smallest Hessian eigenvalues, respectively.²³ McCullough

²¹Section 7.3 in McFadden and Newey (1994) provides an illustrative formula for numerical Hessian approximation. See also Section 9.2.5 in Davidson (2000). The weighted-gradient stopping criterion is largely inspired from ML estimation. See the discussion about stopping criteria in Section 6.3 in Quandt (1983) and in Section 16.5 in Ruud (2000). Figure 16.5 in Ruud provides a very intuitive explanation for the use of the criterion. Dube et al. (2011) derive analytically the elements of the Hessian for the GMM estimation approach employed here in an on-line Appendix.

²²According to Andrews (1997), the best specified definitions of GMM estimators in the literature require one to find a value $\hat{\theta}$ of the estimator that is close to minimizing the criterion function. For example, Pakes and Pollard (1989) require that $\hat{\theta}$ yield a value of the criterion function that is within $o_p(1)$ for consistency and within $o_p(n^{-1})$ for asymptotic normality $\hat{\theta}$. Veall (1990) provides a diagnostic test of the null that a maximum already found is global using results from extreme-value asymptotic theory due to de Haan (1981). Andrews (1997) identifies a serious problem with the power of Veall’s test and proposes a stopping-rule (SR) procedure for the computation of GMM estimators.

²³The condition number of a nonsingular matrix A is given by $\kappa(A) = \|A\| \times \|A^{-1}\|$, where any matrix norm can be used in the definition. In the case of a symmetric positive definite matrix, we have $\|A\| = \lambda_{\max}(H)$ and $\|A^{-1}\| = 1/\lambda_{\min}(H)$. See Section A.2. in Nocedal and Wright (1999).

and Vinod (2003) recommend that solutions for which the Hessian condition number exceeds $1/\sqrt{\epsilon}$, where ϵ is the machine precision should not be accepted uncritically. For MATLAB, $\epsilon = 2.2\text{E}-16$, such that $1/\sqrt{\epsilon} = 6.7\text{E}+07$. Furthermore, for a matrix A , the rough rule of thumb is that as $\kappa(A)$ increases by a factor of 10, you lose one significant digit in the solution of the linear system $Ax = b$ (see Judd (1998), page 68).²⁴

The global minimum at 4.56 corresponding to the 42nd set of starting values for DER2-QN2 implies $\lambda_{max} = 1.65\text{E}+04$ and $\lambda_{min} = 2.84\text{E}-05$, which in turn give $\kappa(H) = 5.80\text{E}+08$. The eigenvector corresponding to the smallest eigenvalue has an extremal element of 0.9986 in the direction of the interaction of price with the log of income. The eigenvector corresponding to the largest eigenvalue has an extremal element of 0.9861 in the direction of the standard deviation term for sugar. This Hessian condition number exceeds the threshold in McCullough and Vinod by almost an order of magnitude.

For automobiles, the implied gradient norm for the estimates in Table A.4 is between 0.05 (DER1-QN1) and 534.13 (STO2-GAL); see panel (b) in Table 3. The weighted-gradient criterion does not exceed 0.002 for the four algorithms that stopped at 178.06, which include DER2-QN2, DER4-SOL, DIR1-SIM and DIR3-GPS. With the exception of DER4-SOL, the Hessians for the remaining algorithms are positive definite and their condition numbers, which do not exceed $8.2\text{E}+03$, are very similar. Therefore, the point at 178.06 may be treated as a local minimum.

The other two terminal points that may be treated as local minima are the ones associated with DER1-QN1 (215.09) and DER3-CGR (215.16). The weighted-gradient criterion for the two algorithms does not exceed 0.15 when rounded to the second decimal point, the Hessians are positive definite and the condition numbers are $2.05\text{E}+03$ and $2.08\text{E}+03$, respectively. Therefore, on the basis of the first- and second-order optimality conditions, the objective function values 178.06, 215.09, and 215.16 correspond to local minima. The condition numbers of the associated Hessians do not exceed the threshold of McCullough and Vinod. In the spirit of our earlier discussion for the cereals, we may call the minimum at 178.06 the “global” minimum although we have not proved or formally tested that we have indeed found a global minimum.

The global minimum at 178.6 associated with the 26th set of starting values for DIR1-SIM implies $\lambda_{max} = 2.4\text{E}+04$ and $\lambda_{min} = 2.9\text{E}+00$, which in turn give $\kappa(H) = 8.03\text{E}+03$.

²⁴Judd argues that a condition number is small if its base-10 logarithm is about 2 or 3 for a computer that carries about 16 significant decimal digits. The implications of ill-conditioning for a wide-class of optimization algorithms can be seen using the iterations in Newton’s method given by $x_{k+1} = x_k - H^{-1}(x_k)g(x_k)$, where H is the Hessian and g is the gradient. In the linear system $Ax = b$, $\kappa(A)$ indicates the maximum effect of a perturbation in b or A in the solution. It can be shown that $\|\delta x\|/\|x\| = \kappa(A) \times \|\delta b\|/\|b\|$ and $\|\delta x\|/\|x + \delta x\| = \kappa(A) \times \|\delta A\|/\|A\|$.

The eigenvector corresponding to the smallest eigenvalue has an extremal element of 0.7753 in the direction of the standard deviation term of HP/WT. The eigenvector corresponding to the largest eigenvalue has an extremal element of -0.9989 in the direction of the standard deviation term for MPG. The Hessian condition number does not seem to raise concerns regarding the numerical precision of the solution, at least based on the metric proposed by McCullough and Vinod.²⁵

To sum up, we provided diagnostics for the analytical gradients and the numerical Hessian of the objective function for the estimates in Tables A.2 and A.4 using tight NFP tolerance. In the case of cereals, the point at which the objective function value is 4.56 satisfies the criteria of a local minimum. Only four of the nine algorithms that converged achieved this minimum. The Hessian condition number may raise some concerns about the numerical precision of the solution based on the metric suggested by McCullough and Vinod (2003). For automobiles, the points at which the objective function is equal to 178.06, 215.09 and 215.16 satisfy the criteria of a local minimum without raising precision concerns based on the same metric. Four of the nine algorithms that converged achieved the first minimum, while a single algorithm achieved the second and third one.

6.3 Additional Local Optima

In this section, we examine whether any terminal points of the nonlinear searches for the various algorithms beyond the ones reported in Tables A.2 through A.4 qualify as local optima. We do so by using the first- and second-order diagnostics that we discussed in the previous section. We first discuss our findings with tight NFP tolerance. We subsequently provide a discussion of our findings with loose NFP tolerance. Before moving to any of these details, we offer a brief motivation for our discussion of the local optima.

Overall, it can be difficult to show that the criterion function of an extremum estimator attains a unique global minimum at the true parameter vector. For example, in nonlinear GMM, conditions for identification are like conditions for unique solutions of nonlinear equations that are known to be difficult (see Section 2.2.3 in McFadden and Newey (1994)). Additionally, it is often challenging to find the unique global minimum of a criterion function setting aside the case of a globally convex criterion function where there can be at most one local minimum that is also the global minimum. As a result, two consistency theorems for extremum estimators are available, one for a global optimum and one for a local optimum.²⁶

²⁵Additional diagnostics, related to the validity of the quadratic approximation at the terminal point of the nonlinear search, at least for the derivative-based algorithms, which have been suggested in the literature include the profile t-plot and plots of the profile traces (e.g., see Bates and Watts (1988)). These diagnostics are readily available as part of some optimization routines; e.g., see the MAXLIK routine of GAUSS.

²⁶See Theorems 4.1.1 and 4.1.2 in Amemiya (1985) or Theorems 5.1 and 5.2 in Cameron and Trivedi (2005).

When there is more than one local optima, the consistency theorem for a local optimum states that one of the local optima is consistent, but provides no guidance to which one is the consistent.²⁷ Therefore, although any of the plausible, from economic-theoretic viewpoint, local optima can be a consistent root, the studies in Table 1 seem to not have examined the possibility of such local optima. At least, they don't discuss such a possibility explicitly.²⁸ As we mentioned in the previous section, we are not aware of a study formally testing that has indeed found a global minimum following, say, the procedure in Andrews (1997), either.

In the case of cereals, we did not identify any local minima that were different from the one at 4.56 with tight NFP tolerance. For automobiles, 49 terminal points of the nonlinear searches indicate an objective function value equal to 215.09 when rounded to the second decimal point.²⁹ The vast majority of these points are associated with DER4-SOL and DIR1-SIM; 23 and 15, respectively. The associated parameter estimates are qualitatively very similar across these 49 points. The maximum gradient norm across the 49 terminal points is 0.1, the maximum weighted gradient criterion is 0.0002 and the implied Hessians are all positive definite. For the same set of local minima, the minimum normalized eigenvalue, $\lambda_{min}/\lambda_{max}$, falls within a very tight range, namely, 4.79E−04 to 4.89E−04. The range for $\kappa(H)$ is 2.04E+03 to 2.09E+03. We also identified additional local optima with objective function values of 207.72, 216.03, and 226.94. Across all these three local optima, the maximum of the implied gradient norms does not exceed 0.09, the weighted-gradient criterion is zero (to the fourth decimal point) and the implied Hessians are all positive definite. The minimum normalized eigenvalues are between 2.39E−05 and 2.14E−04 and $\kappa(H)$ lies between 4.68E+03 and 4.19E+04.

Our results using a loose NFP tolerance underscore the findings in Dube et al. (2011). In the case of cereals with loose NFP tolerance, the DER5-KNI algorithm hovered around an objective function value of about 15.5. More precisely, 16 sets of starting values gave rise

Amemiya (page 230) states that identification is synonymous to “the existence of a consistent estimator.”

²⁷Cameron and Trivedi (page 127) argue that it is best in such cases to consider the global optimum and apply their Theorem 5.1. According to McFadden and Newey (page 2117), as long as the extremum estimator is consistent and the true parameter is an element of the interior of the parameter space, an extremum estimator will be a root of the first order conditions asymptotically, and hence, will be included among the local optima. Amemiya (page 111) suggests two ways to gain some confidence that a local optimum is a consistent root. First, the solution gives a reasonable value from an economic-theoretic viewpoint. Second, the iteration by which the local optimum was obtained started from a consistent estimator.

²⁸The only exception seems to be footnote 26 in Berry et al. (1999), which reads as follows: “There is also the issue of the shape of our objective function, in particular the presence of local minima, and the ability of our numerical procedures (which includes a choice of starting values and of stopping tolerances) to find its overall minimum. We experimented with alternate starting values and tolerances and sometimes found the minimization algorithm stopping at local minima that were slightly different than the overall minima reported in the text.”

²⁹Our count excludes the point for DER1-QN1 in Table A.4.

to function values between 15.46 and 15.60 when rounded to the second decimal point. The coefficient estimates implied by these 16 sets of starting values are largely identical with the exception of the two coefficients associated with the interaction of price with log income and log income squared. The range of these two coefficients is -1.42 to 2.00, and 0.09 to 0.27, respectively.

In all 16 instances, although the gradient norm (weighted-gradient criterion) is between 0.0393 (-0.0002) and 0.0968 (0.0007), the Hessians exhibit both positive and negative eigenvalues. These diagnostics are consistent with a *saddle point*.³⁰ The range of the condition number $\kappa(H)$ is between $-1.50\text{E}+03$ and $-2.75\text{E}+02$ across the 16 points. Hence, the numerical accuracy of the solutions does not seem to be a concern. In the case of automobiles with loose NFP tolerance, we did not identify any local minima beyond the ones associated with the smallest function value in Table A.3.³¹

7 Implications for Economic Variables of Interest

7.1 Preliminaries

In this section, we examine the implications of the variation in the terminal points of the nonlinear searches for economic variables of interest, which are routinely studied in the literature, such as elasticities, consumer welfare, and firm profits. Following our extensive optimization exercise, the number of observations for the analysis of the variation in such variables, especially at the product level, is rather immense. Every combination of parameter starting values, optimization algorithm, and NFP tolerance yields an elasticity matrix for each market; of course, not all of these combinations are meaningful. In the case of the cereal data, there are 94 markets with 24 products in each market leading to 2,256 market and product combinations. In the case of the automobile data, there are 20 markets, with the number of products in each market being between 72 and 150 leading to 2,217 product and market combinations.

At the product level, we focus on the implication for the own-price elasticities for two products in each dataset. The first is the product with the largest observed quantity sold

³⁰More specifically, KNITRO generated the exit code of 0 (-100) in 14 (2) instances. According to Appendix A in [Waltz and Plantenga \(2009\)](#), the exit code 0 indicates that “The final solution satisfies the termination conditions for verifying optimality,” while exit codes -100 to -199 indicate that “A feasible approximate solution was found.”

³¹Recently, [Judd and Skrainka \(2011\)](#) show that Monte Carlo integration for evaluating the market share integrals creates “ripples” in the surface of the objective function which generate spurious local maxima. [Skrainka \(2011\)](#) argues that when instruments are highly collinear, which is often the case for the BLP-type instruments based on product characteristics, the GMM weighting matrix has a high condition number and the outer-loop nonlinear solver finds many local optima.

(top product). The second is the product with the median observed quantity sold (median product). In the case of cereals (automobiles), we work with 240 (387) observations for each of the two products of interest. For cereals, the top product is brand 5 (a Kellogg’s brand) that appears in market 53 and its market share is 45%. The median product is brand 13 (a General Mills brand) that appears in market 89 and its share is 1%. In the case of automobiles, the top product appears in market 2 (1972) and its market share is about 1%. The median product appears in market 16 (1986) and its market share is 0.05%.³²

At the market level, we examine the implications for aggregate elasticity, as well as the change in profits and consumer welfare for two hypothetical mergers. We calculated the aggregate elasticity by simulating a 1% price increase for all products. We use compensating variation as a measure of change in consumer welfare. In the case of the cereal data set, we assume Kellogg’s and General Mills merge. For the automobile data, we assume GM and Chrysler merge. As we discussed earlier, we use the approximate solutions for the post-merger prices of Equation 13 in Section 3. Similar to the analysis at the product level, we work with 240 observations in the case of cereals and 387 observations in the case of automobiles.

In the two subsequent sections, we limit our attention to results implied by those sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance excluding some additional sets of results. First, we excluded all the results implied by combinations of starting values and optimization algorithms that gave rise to problematic pre- and post-merger market shares (e.g., NaNs). Second, in the case of cereals (automobiles), we excluded results associated with combinations of starting values and optimization algorithms that gave rise to GMM objective function values in excess of 134.92 (401.05).³³ The process just described gave rise to 240 observations in the case of cereals and 387 observations in the case of automobiles.

In general, we find that the variation across the set of candidate estimates for cereals is much smaller than the variation in the automobile data. This is not surprising given that three of the algorithms nearly always converge at the GMM objective value of 4.56 for cereals. However, it is important to keep in mind that this point is found only for these three algorithms. Therefore, if a researcher were to rely only on the other eight algorithms, the variation would be more pronounced.

³²The products with the largest and median observed quantity sold are identified by the NEWMODV data field as BKRIVE72 and NIPULS84, respectively.

³³In the case of cereals, 134.92 corresponds to the 75th percentile of the objective function value distribution. The 90th percentile is 314.58 and the maximum is 1,904.78. For automobiles, 401.05 is the 95th percentile of the objective function value distribution. The 99th percentile is 4,668.81 and the maximum is in excess of 12,000.

7.2 Product Level

Panel (a) in Figure 2 provides a histogram of the own-price elasticity (own elasticity, henceforth) for the top product in cereals. The own elasticity across the 240 observations is between -2.47 and -1.34 with an average of -1.98 and a standard deviation of 0.14. Using the absolute value of the mean, the implied coefficient of variation (CoV), a unit-free measure of dispersion, is equal to 0.07.³⁴ Therefore, we see notable variation, but given the number of routines that terminated at an objective value of 4.56 the variation is not extreme. The median own elasticity (-1.99) is essentially identical to the value of the own elasticity associated with the smallest objective function value of 4.56.

We provide a histogram of the own elasticity for the cereal median product in panel (a) of Figure 3. The own elasticity across the 240 observations is between -5.44 and -2.59 with an average of -3.49 and a standard deviation of 0.73. The implied CoV is 0.21. The median is -3.47. Similar to the largest product, the variation is notable. The 75h percentile of the own elasticity distribution (-2.62) corresponds to the value of the own elasticity associated with the smallest objective function value of 4.56.

Further support for the variation in own elasticities due to combinations of starting values and algorithms is provided by the histogram in panel (a) of Figure 4. This histogram is based on coefficients of variation (CoV) of the own elasticity for 2,144 product-market combinations across all 240 combinations of optimization algorithms and starting values.³⁵ That is, each CoV is calculated using 240 observations. If there were no variation in the own elasticities the CoV should be zero. The CoV distribution has a mean of 0.08 that is indistinguishable from the median.

For the own elasticity of the top automobile product, we see values as low as -4.72 and as high as -0.51, if we exclude 4 values exceeding zero—see panel (b) in Figure 2. The mean across the 383 (387-4) observations is -2.85, which is almost identical to the median, and the standard deviation is 0.61; the CoV of 0.21. Similar to cereals, we see substantial variation in the distribution of the own elasticities with a rather prominent spike around -3.5. The variation is notable even when we limit our attention to the 22 observations associated with the local optima. Recall from our earlier discussion that we identified 5 local optima in the case of the automobile data with objective function values of 178.06, 207.72, 215.09, 216.03, and 226.94. The implied own elasticities are -2.64, 1.46, -3.41, -1.86, and -2.76, respectively. The own elasticity associated with the smallest objective function value of -178.06 is slightly below the 75 percentile that is equal to -2.39. Notice also the difference in the own elasticities

³⁴Throughout this section, we calculate the CoV using the absolute value of the mean.

³⁵The number of product-market combinations is less than 2,256 because we exclude observations exceeding the 95th percentile of the CoV distribution, which is 0.175.

(-3.41 vs. -1.86) for the two local optima with very similar objective function values, namely, 215.09 and

The histogram of own elasticities for the median automobile product is available in panel (b) of Figure 3. In this case, the minimum across the 387 observations is smaller than the maximum by a factor of 10; -9.73 vs. -0.93. Once again, we observe substantial variation in the own-elasticity distribution with a prominent spike in the neighborhood of -3.5. The mean is -2.85 and the median is -2.92. The standard deviation is 0.81 and the implied CoV is 0.28. To account for the skewness in the distribution, we constructed the histogram excluding own-elasticity values below -3.82, the 5th percentile across the 387 observations. Similar to the largest product, the variation is substantial when we limit our attention to the observations associated with the five local optima: -3.88 (178.06), -3.08 (207.72), -3.41 (215.09), -2.41 (216.03), -2.30 (226.94). The own elasticity associated with the smallest objective function value of -178.06 may be viewed as an outlier given that the 5th percentile is equal to -3.82. At the same time, the two close local optima at 215.09 and 216.03 give rise to notably different own elasticities.

Finally, we constructed the CoV histogram across 2,107 automobile product-market combinations for each of the 387 combinations of optimization algorithms and starting values—see panel (b) of Figure 4.³⁶ Therefore, each CoV is calculated using 387 observations. As we can see, the hump of the distribution covers the 0.20-0.25 range and a long right tail is prominent. The mean and median of the CoV distribution are very similar, 0.25 and 0.23, respectively. Similar to cereals, and to a larger extent, we see substantial variation in the own-elasticity distribution due to combinations of algorithms and starting values, at least, as captured by the CoV.

7.3 Market Level

We start our discussion in this section with the histogram of the average aggregate elasticity for cereals in panel (a) of Figure 5. More specifically, for each of the 240 sets of parameter estimates, we first calculated an aggregate elasticity for each of the 94 markets simulating a 1% price increase for all brands. We then took a quantity-weighted average across the 94 markets. Given that the cereal data do not contain information about the market size, we assumed a total market size (including the outside good) of 250 million servings per day. When expressed in million servings per day, the quantity weight for each market is equal to

³⁶The number of product-market combinations is less than 2,217 because we exclude observations exceeding the 95th percentile of the CoV distribution, which is 0.67.

the share of the inside goods times 250.³⁷

We see values for aggregate elasticity between -1.78 and -0.84 with the distribution having a prominent peak at about -1.3, which is very close to the mean and the median. This is also the value of the average aggregate elasticity if we limit our attention to the minimum with objective function value of 4.56. The standard deviation is 0.11. Given that the 95th percentile is -1.25, we would conclude that the average market is inelastic only for a handful of estimates. The variation is less pronounced if we exclude observations in the top and bottom 5% of the distribution—in this case, the elasticity values are between -1.56 and -1.25.

We also calculated a quantity-weighted average annual change in profits following a hypothetical merger between Kellogg’s and General Mills. As we discussed earlier, we approximated the post-merger prices using Equation 13 in Section 3. The histogram of such profit changes is available in panel (a) of Figure 6. In a little more detail, we calculated the annual change in profits in each market multiplying the market size by 365, and, hence, it is expressed in millions of dollars. We then took a quantity-weighted average across the 94 markets for each of the 240 sets of estimates. Similar to the histogram for the average aggregate elasticity, we don’t see a single mass point—actually, we observe substantial variation. The average annual change in profits due to the merger ranges from \$104.3m to \$229.7m with a mean (median) of \$170.8m (\$177.1m) and standard deviation of \$15.7m. When we limit our attention to the estimates that give rise to the objective function value of 4.56, the change in profits is about \$177.1m. The variation in profit change when we exclude the top and bottom 5% of the distribution is still substantial, \$138.3m to \$182.1m.

Furthermore, we calculated a quantity-weighted average annual change in consumer welfare for cereals due to the hypothetical merger using compensating variation. For each of the 94 markets, we first calculated an average compensating variation (ACV) using Equation 14 in Section 3. We then calculated an annual market compensating variation (MCV) in millions of dollars as $ACV \times 250 \times 365$. Finally, we calculated an average across the 94 markets using the units of the inside goods as weights for each of 240 sets of estimates. The histogram of the change in consumer welfare is available in panel (a) of Figure 7. As it was the case with the aggregate elasticity and change in profits, we see substantial variation in MCV. Across the 240 estimates, the range is from -\$975m to -\$469m with a mean (median) of -\$671m (-\$651m) and standard deviation of approximately \$60m. The change in consumer welfare corresponding to the objective function value of 4.56 is almost identical to the

³⁷The U.S. population was approximately 250 million based on Census figures for July, 1990; see <http://www.census.gov/popest/data/national/totals/1990s/tables/nat-agesex.txt>. The assumption about the total market size is admittedly somewhat arbitrary. However, it is inconsequential for the variation of the results reported here because the market size operates as a scaling factor.

median. Even after excluding the top and bottom 5% of the MCV distribution, the MCV exhibits substantial variation, namely, -\$744m to -\$569m.

The aggregate-elasticity histogram for the automobile data is plotted in panel (b) of Figure 5. This histogram, as well as the histograms for the change in profit and the compensating variation, exclude market 6 (1975) due to its extraneous observations. We calculated the aggregate elasticity following the simulation approach we described for cereal.³⁸ The aggregate elasticity ranges from -1.73 to -0.52 with a mean (median) of -1.23 (-1.29) and a standard deviation of 0.21. The ratio of the minimum to the maximum is about 3 in absolute value and the aggregate elasticity exceeds -1 in 59 instances. The variation is still substantial when we exclude the top and bottom 5% of the aggregate-elasticity distribution, that is values below -1.52 and above -0.83. Limiting our attention to the aggregate elasticity values associated with the local optima, we get the following: 178.06 (-0.64), 207.72 (-1.03), 215.09 (-1.38), 216.03 (-0.96), and 226.94 (-1.29).

The distribution of the quantity-weighted average change in profits due to a hypothetical merger between GM and Chrysler is available in panel (b) of Figure 6. Excluding the top 5% and bottom 10% of the distribution, which corresponds to discarding values below \$302m and above \$2,287m, leaves us with 330 observations. Following the trimming of the distribution just described, we see values between \$302m and \$2,287 with a mean (median) of \$782m (\$715m) and a standard deviation of \$343m. The change in profits associated with the various local optima are as follows: 178.06 (\$2,289m), 207.72 (\$873m), 215.09 (\$755m), 216.03 (\$1,567), 226.94 (\$580m).

Finally, we provide the histogram of the quantity-weighted average market compensating variation (MCV); see panel (b) in Figure 7. We calculated MCV as the product of the average compensating variation and the total market size. After dropping the observations in the bottom 5% of the distribution (values below -\$4,321m) or exceeding zero, which leads to 364 observations, we see values between -\$4,321m and -\$542m. The mean (median) is -\$2,178m (-\$2,008m) and the standard deviation is \$627m. The change in consumer welfare captured by MCV at the various local optima is 178.06 (-\$4,307m), 207.72 (-\$2,590m), 215.09 (-\$2,008m), 216.03(-\$3,279m), and 226.94 (-\$1.829m).

Overall, the distributions of the automobile economic variables of interest at the market level indicate that the values of such variables associated with the smallest objective function value of 178.06 are outliers. In addition, parameter estimates that lead to very similar objective function values can have extremely different economic predictions. For

³⁸The automobile data contain information about the total market size—i.e., including the outside good. We constructed our quantity-weighted average across the 19 markets for each of the 387 sets of estimates using the units associated with all inside goods for all three economic variables of interest.

example, comparing the two local minima at 215.09 and 216.03, the change in profits is over two times larger for the parameter values associated with 216.03, while the change in consumer welfare is over 50 percent larger. This underscores our discussion of the potential for a “horse race” even when the researcher is convinced that the global minimum is the consistent root. One could imagine a situation where, for a slightly different sample, the parameter values associated with the minimum at 216.03 yield a smaller objective value than the parameter values associated with the minimum at 215.09.

8 Conclusions

Empirical industrial organization has been increasingly relying on highly nonlinear structural models, and, probably, more so compared to other neighboring fields, such as public and labor economics. The reasons for such divergence have been discussed recently in a rather lively manner in [Angrist and Pischke \(2010\)](#) and [Nevo and Whinston \(2010\)](#). At the same time, a prominent class of such models, namely, the demand models for differentiated products of the [Berry et al. \(1995\)](#) tradition, are becoming increasingly popular in trade, education, housing, health, and environmental economics shedding light on a wide spectrum of important economic questions.

Nonlinear models are very often synonymous to an objective function that is not globally concave or convex. Obtaining parameter estimates and performing inferences is possible, in principle, using a nonlinear search algorithm along with a set of multiple starting values and stopping rules. Both exercises, however, can be particularly challenging when it comes to implementation. In this paper, we document some of the challenges we experienced estimating BLP-type demand models using two widely known datasets, numerous search algorithms a large number of starting values, and varying tolerances of the fixed-point iterations that allow the researcher to infer the structural econometric error.

Our findings point to instances of convergence at points where the first- and second-order optimality conditions fail. We also find that various combinations of optimization algorithms, starting values, and fixed point-iterations may lead to convergence at multiple local optima, as well as to instances of convergence failure. Even upon convergence under tight tolerance for the fixed-point iterations, we find substantial variation in the objective function value both within and across optimization algorithms. Although derivative-based algorithms seem to perform better in the case of the cereal dataset, there seems to be a tight-race between derivative-based and direct-search algorithms for the automobile dataset. This variation goes in hand with variation in parameter estimates and translates into variation of the demand models’ economic predictions, such as price elasticities, consumer welfare, and firm profits.

In the cereal dataset, the value of the own-price elasticity of the product with the largest market share exhibits a range such that the smallest and largest values differ by a factor of 2 depending on the combination of optimization algorithm and starting values. At the market level, the range is qualitatively similar for the aggregate elasticity, as well as for the change in consumer welfare and firm profits following a hypothetical merger in the industry.

In the automobile dataset, the range of the own-price elasticity for the product with the highest market share is such that the smallest to the largest value differ by a factor of 8. The same factor is about 3.5 in the case of the aggregate elasticity and close to 7 for the change in firm profits due to a hypothetical merger. The range for the change in the consumer welfare has implications that are similar to that of the change in firm profits. All of these economic variables of interest exhibit notable variation across multiple optima that we identify through a thorough review of first- and second-order optimality conditions.

Based on our experience, we advocate the use of multiple optimization algorithms with a large number of starting values along with a thorough review of the algorithms' solution paths and a careful examination of the first- and second-order optimality conditions. We also support the discussion of any economically plausible local optima that may emerge in the course of the optimization exercise. As a result, we are encouraged by the comprehensive summary of the optimization exercise in [Goldberg and Hellerstein \(2011\)](#). Encouraging messages come also from recent papers, such as [Dube et al. \(2011\)](#), [Judd and Skrainka \(2011\)](#), and [Skrainka \(2011\)](#), that push the frontier in the estimation of BLP-type demand models and contribute to a better understanding of some of the issues we have identified in this work. To date, Goldberg and Hellerstein provide the best example of how empirical researchers employing structural econometric modeling should receive the message that we try to convey in this paper.

A Appendix

A.1 Parameter Estimates

Focusing on the performance of the optimization algorithms and NFP tolerances only in terms of the value of the objective function can be misleading. If the objective function is steep around the true parameter values, a nonlinear search may yield parameter values that are close to the true values, but have an objective function value that is very different. Alternatively, very different parameter estimates may lead to very similar objective function values. From an empiricist’s point of view what ultimately matters is how the variation of the objective function value and the associated parameter estimates translates into variation of economic variables of interest—this is the subject of our discussion in Section 7. In this Appendix, we briefly discuss variation in the parameter estimates.

We present parameter estimates associated with the minimum value of the objective function across the various starting values for each optimization algorithm in Tables A.1 and A.2 for cereals. The difference between the two tables is that the NFP tolerance is loose in the former but tight in the latter. The results for automobiles appear in Tables A.3 and A.4. Our discussion of the results in these four tables will exclude pairs of algorithms and starting values that failed to converge. In the case of cereals, we exclude DIR2-MAD and STO1-SIA under both loose and tight NFP tolerance because they stopped by exceeding the maximum number of function evaluations. For automobiles, we exclude STO1-SIA under both loose and tight NFP tolerance, DIR1-SIM under loose NFP tolerance, and DIR2-MAD under tight NFP tolerance for the same reason. As a result, our discussion below is based on 18 sets of parameter estimates for both data sets.

For the cereal data with loose NFP tolerance, as Table A.1 illustrates, there is still a lot of variation in the objective function values; from 4.56 (DER2-QN2, DER4-SOL and DER5-KNI) to 108.13 (STO3-SIG). This should not come as a surprise to the reader given our discussion of the BaW plots earlier in this section. The value of the mean price coefficient lies between -62.74 (DER2-QN2) and -30.23 (DER1-QN1). The standard-deviation term for price ranges from 1.06 (DIR3-GPS) to 3.31 for (DER4-SOL). For the remaining standard-deviation terms, the maximum absolute value of the associated coefficient may exceed the minimum value by a factor of almost 40. For example, this is the case for sugar content, the minimum value is 0.003 (DIR1-SIM) and the maximum value is 0.12 (STO3-SIG). For the terms related to interactions with demographics, the widest range is that for the interaction of price with the log of income: -5.32 (STO3-SIG) to 588.48 (DER2-QN2).

Implementing a tight NFP tolerance has a small impact on the variation in objective function values as we can see in Table A.2. Their range remains almost identical to that

under loose NFP tolerance; 4.56 (DER2-QN2, DER4-SOL and DER5-KNI) to 108.13 (STO3-SIG). The range of the values of the mean price coefficient also remains largely unchanged; -62.74 (DER5-KNI) to -29.4 (DIR1-SIM). In the case of the standard-deviation term for price, the range is wider relative to its loose-NFP counterpart, 0.73 (STO2-GAL) to 3.31 (DER4-SOL). For the remaining standard deviation terms, the minimum value of the constant and the maximum value of mushy both decrease as we move from the loose to the tight NFP tolerance. The range for sugar content and the terms related to the interactions with demographics remain qualitatively similar to those under loose NFP tolerance.

The loose NFP tolerance for automobiles leads to objective function values between 198.20 (DER2-QN2) and 277.62 (DER5-KNI) in Table A.3 even when we focus on the set of starting values that generate the smallest objective function values within each optimization algorithm. The mean price coefficient lies between -0.55 (DER2-QN2) and -0.36 (DER5-KNI). The remaining mean coefficients also exhibit significant variation. For example, the range of the HP/WT coefficient is -1.31 to 3.07, while the range of the MPG coefficient is -4.06 to 0.02. The absolute value of the standard-deviation term for price exhibits a range of 0.14 (DER5-KNI) to 0.29 (DER2-QN2). The remaining standard-deviation terms exhibit wider ranges. For example, the range of the HP/WT coefficient is 0.12 to 6.24, while the one for A/C is 0.4 to 7.8.

The tight NFP tolerance leads to three algorithms achieving the same minimum objective function value of 178.06. The range of the mean price coefficient is qualitatively similar to that under loose NFP: -0.46 (DER1-QN1) to -0.28 (STO2-GAL). The most notable changes for the remaining mean coefficients are those for the constant (minimum value of -9.75 vs. -14.69) and AC (minimum value of 0.44 vs. -5.15). The most notable change in the standard deviation coefficients is that for AC, where the upper end of its range decreases from 7.82 to 1.89 as we move from the loose to the tight NFP tolerance. The range of the standard deviation coefficient for price is now 0.09-0.18 as opposed to 0.14-0.29 with loose tolerance.

To sum up, the variation in the objective function value documented in the previous section goes in hand with variation in the parameter estimates documented in this section. The variation in the parameter estimates remains when we limit our attention to the estimates that correspond to the minimum objective function value for each optimization algorithm under the two alternative NFP tolerances considered. For example, in the case of cereals, the mean price coefficient is as low as -60 and as high as -30. The standard deviation price coefficient is between 0.7 and 3. For automobiles, the mean price coefficient is roughly between -0.6 and -0.3 and the standard deviation coefficient is 0.1 and 0.3. This variation is qualitatively similar under both loose and tight NFP tolerance.

A.2 Additional Optimization Exercises

We repeated our optimization exercises for the derivative-based algorithms using a tolerance for the change in the parameter vector and the objective function value of $1\text{E-}06$ as opposed to $1\text{E-}03$. We also used a tight NFP tolerance maintaining the maximum number of function evaluations (4,000) and NFP iterations (2,500). As we discuss below, only DER3-CGR reached the maximum number of function evaluations. It did so 32 times in the case of cereals and 10 times in the case of automobiles.

For cereals, DER4-SOL converged to the smallest objective function value of 4.56 for all 50 sets of starting values. DER5-KNI converged to the same function value 40 times when we imposed a lower bound of zeros on the parameter vector and 34 times when we did not impose such a lower bound. Across all these 74 instances, DER5-KNI produced 69 times an exit code of 0 and 5 times an exit code of -100. Additionally, DER1-QN1 (DER2-QN2) converged to the smallest objective function value 26 (31) times. Although DER3-CGR never converged to the smallest objective function value, it hovered around objective function values between 15.4 and 15.8 reaching the maximum number of function evaluations 24 times. These objective function values are very close to the objective function values associated with indefinite Hessians that we discussed in Section 6.3.

Figure A.1 provides the distribution of the objective function values for the derivative-based algorithms with outer tolerance of $1\text{E-}03$ (panel a) and outer tolerance of $1\text{E-}06$ (panel b) for cereals. As we did in the case of Figure 1, we limit our attention to those sets of starting values that allowed the algorithms to converge under tight tolerance and gave rise to values not exceeding 110. Imposing a tight outer tolerance reduces substantially the variation in the distribution of the objective function values—about 90% of them are equal to the smallest objective function value of 4.56.

For automobiles, DER5-KNI converged 7 times to an objective function value of 178.06 when we did not impose a lower bound of zeros on the parameter vector generating exit codes 0 (1 time), -100 (2 times) and -101 (4 times). DER1-QN1, DER2-QN2, and DER4-SOL, all converged to the objective function of 178.06 value once. Recall from our discussion in the main text that DER5-KNI did not converge to the objective function value of 178.06 when we imposed a lower bound of zeros. Additionally, about 150 sets of the starting values for the non-KNITRO algorithms and DER5-KNI (unbounded) led to convergence with an objective function value of 215.09 (rounded to the second decimal point). Furthermore, 65 sets of starting values for the same algorithms led to convergence with an objective function value of 226.94 (rounded to the second decimal point). The exit codes for DER5-KNI (unbounded) at 226.94 were 0 (11 times) and -100 (1 time). DER5-KNI with a lower bound of zeros on the parameter vector converged 39 times with an objective function value of 277.73 (rounded

to the second decimal point) and an exit code of 0. DER3-CGR was the only algorithm that reached the maximum number of function evaluations (10 times).

Figure A.2 provides the distribution of the objective function values for the derivative-based algorithms with outer tolerance of 1E-03 (panel a) and outer tolerance of 1E-06 (panel b) for automobiles. Similar to Figure 1, we limit our attention to those sets of starting values that allowed the algorithms to converge under tight NFP tolerance and gave rise to objective function values not exceeding 430. Imposing an outer tolerance of 1E-06 gives rise to a distribution that is similar to the distribution under 1E-03. The main implication of the tighter outer tolerance is to shift some mass of the distribution to the local optima of 215.09 and 226.94.

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Table 1: Papers using BLP-type demand models with aggregate data in selected journals

Journal	Paper
American Economic Review	Berry, Levinsohn and Pakes (1999)
Econometrica	Berry, Levinsohn and Pakes (1995)
	Goeree (2008)
	Nevo (2001)
Journal of Political Economy	Berry, Levinsohn and Pakes (2004)
	Petrin (2002)
Review of Economics and Statistics	Nevo (2003)
	Rekkas (2007)
Review of Economic Studies	Berry, Levinsohn and Pakes (2004)
	Nakamura and Zerom (2010)
	Villas-Boas (2007)
Rand Journal of Economics	Akerberg and Rysman (2005)
	Armantier and Richard (2008)
	Berry (1994)
	Bonnet and Dubois (2010)
	Chu (2010)
	Copeland (2011)
	Davis (2006)
	Iizuka (2007)
	Nevo (2007)
	Song (2007)
	Villas-Boas (2009)

Note: The list of journals is limited to the leading general interest and industrial organization journals. In addition, only papers that contain the main ingredients of the BLP approach for the estimation of random-coefficient Logit models using aggregated data as discussed in Section 1 are included.

Table 2: Optimization Algorithms

Class	Description	Source	Acronym	Routine	Documentation
Derivative-based	Quasi-Newton 1	MathWorks	DER1-QN1	<i>fminunc.m</i>	http://www.mathworks.com/help/toolbox/optim/ug/fminunc.html
	Quasi-Newton 2	MathWorks	DER2-QN2	<i>ucminf.m</i>	http://www2.imm.dtu.dk/~hbn/Software/\#GO
	Conjugate Gradient	Pub. Available	DER3-CGR	<i>conj_grad.m</i>	http://www2.imm.dtu.dk/~hbn/Software/\#GO
	SOLVOPT	Pub. Available	DER4-SOL	<i>slopt.m</i>	http://www.kfunigraz.ac.at/imawww/kuntsevich/solvopt/
	KNITRO	Ziena Optimization	DER5-KNI	<i>ktrlink.m</i>	http://www.mathworks.com/help/toolbox/optim/ug/ktrlink.html
Deterministic Direct Search	Simplex	MathWorks	DIR1-SIM	<i>fminsearch.m</i>	http://www.mathworks.com/help/techdoc/ref/fminsearch.html
	Mesh Adaptive Direct Search	MathWorks	DIR2-MAD	<i>patternsearch.m</i>	http://www.mathworks.com/help/toolbox/gads/patternsearch.html
	Generalized Pattern Search	MathWorks	DIR3-GPS	<i>patternsearch.m</i>	http://www.mathworks.com/help/toolbox/gads/patternsearch.html
Stochastic Direct Search	Simulated Annealing	Pub. Available	STO1-SIA	<i>sa.m</i>	http://www.stanford.edu/~doubleh/eco273/sa.txt
	Genetic Algorithm GADS	MathWorks	STO2-GAL	<i>ga.m</i>	http://www.mathworks.com/help/toolbox/gads/ga.html
	Simulated Annealing GADS	MathWorks	STO3-SIG	<i>simulannealbnd.m</i>	http://www.mathworks.com/help/toolbox/gads/simulannealbnd.html

Table 3: Gradient and Hessian diagnostics by optimization algorithm using tight NFP tolerance

Algorithm	Obj. Fun. Value	$\ g\ _\infty$	$g'H^{-1}g$	$\kappa(H)$
DER1-QN1	19.554	1.088	26.747	3.97E+07 *
DER2-QN2	4.562	0.230	0.000	5.80E+08 *
DER3-CGR	19.929	13.231	39.737	2.88E+07 *
DER4-SOL	4.562	0.009	0.000	5.76E+08 *
DER5-KNI	4.562	0.041	0.000	5.82E+08 *
DIR1-SIM	17.549	7.787	15.865	1.00E+08 *
DIR2-MAD	16.055	17.302	15.513	1.51E+08 *
DIR3-GPS	50.993	472.767	34.194	8.72E+06 *
STO1-SIA	131.320	493.765	365.304	2.53E+07 *
STO2-GAL	31.594	86.113	25.822	-2.25E+06
STO3-SIG	108.130	412.090	-2.850	-9.61E+01

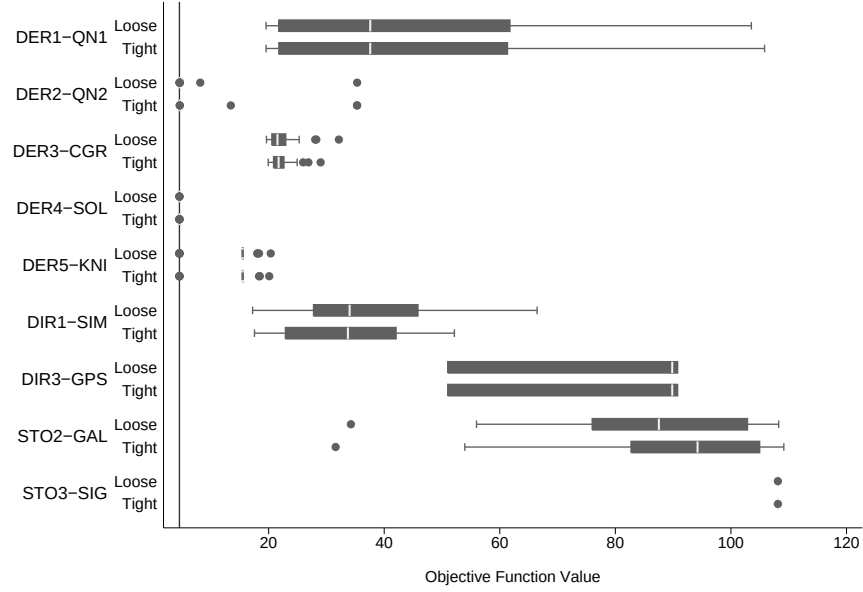
(a) Cereals

Algorithm	Obj. Fun. Value	$\ g\ _\infty$	$g'H^{-1}g$	$\kappa(H)$
DER1-QN1	215.091	0.048	0.000	2.05E+03 *
DER2-QN2	178.061	5.034	0.002	8.22E+03 *
DER3-CGR	215.163	4.882	0.148	2.08E+03 *
DER4-SOL	178.060	0.148	0.000	-2.23E+03
DER5-KNI	277.733	17.498	-17.476	-9.02E+02
DIR1-SIM	178.060	0.328	0.000	8.03E+03 *
DIR2-MAD	208.064	27.159	0.782	3.23E+04 *
DIR3-GPS	178.060	0.070	0.000	7.32E+03 *
STO1-SIA	191.194	527.290	38.985	1.16E+04 *
STO2-GAL	205.434	534.134	56.750	4.68E+03 *
STO3-SIG	186.234	97.869	20.599	4.52E+03 *

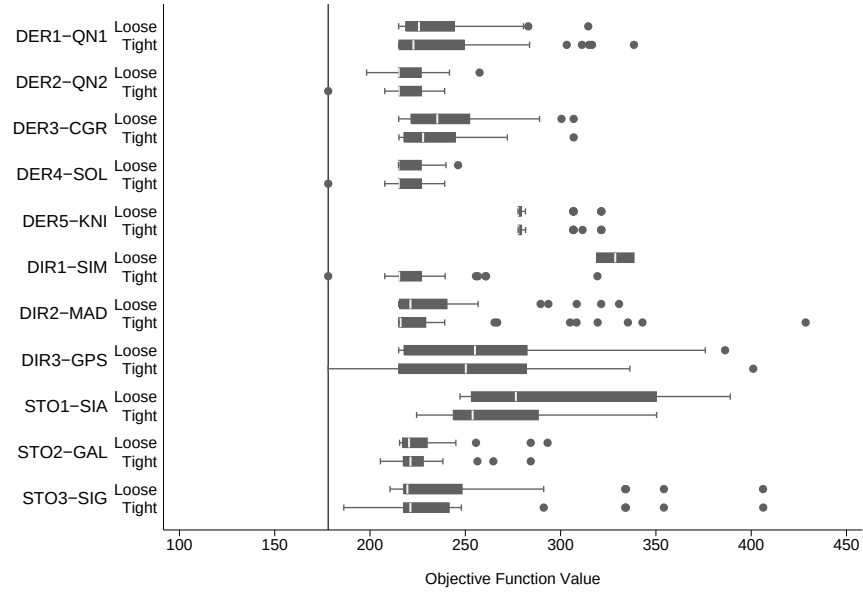
(b) Automobiles

Note: The naming convention of the optimization algorithms of the “Algorithm” column follows Table 2. We use g and H to denote the gradient and the Hessian of the objective function value evaluated at the parameter estimates in Table A.2 for cereals and in Table A.4 for automobiles. We use $\kappa(H)$ to denote the Hessian condition number. The asterisk (*) indicates a positive definite Hessian matrix.

Figure 1: Box-and-Whisker plots of the objective function value by optimization algorithm and NFP tolerance for cereals and automobiles



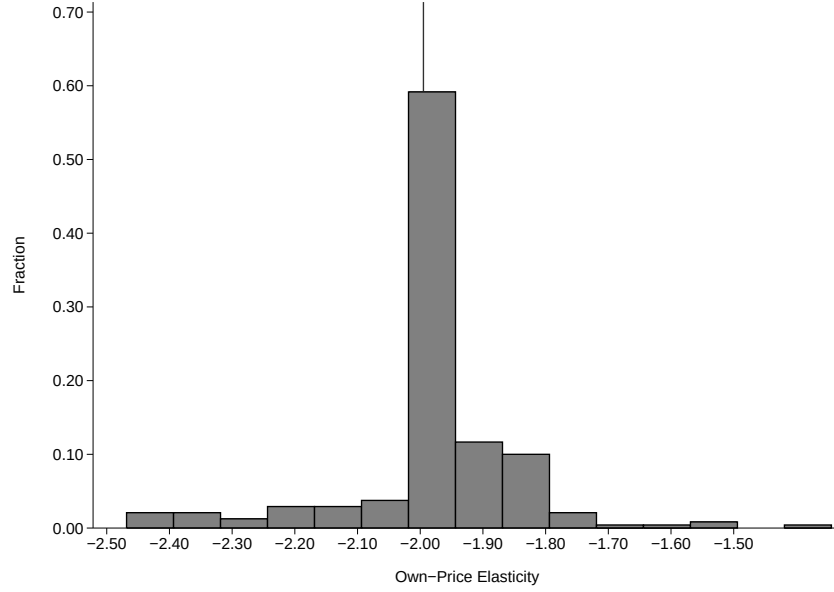
(a) Cereals



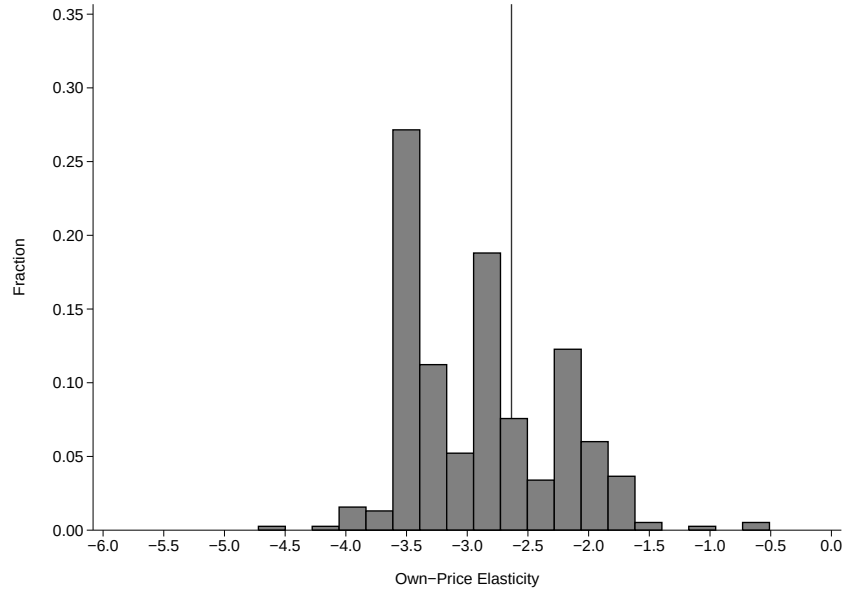
(b) Automobiles

Note: The naming convention of the optimization algorithms on the vertical axis of the Box-and-Whisker plots (BaW) in this figure follows Table 2. Loose and Tight refer to the alternative nested fixed-point (NFP) tolerances described in Section 4. The vertical solid line indicates the minimum objective function value for the combinations of optimization algorithms, NFP tolerances, and starting values that converged. We implemented DER5-KNI using a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search.

Figure 2: Own-price elasticity histogram of the top product with tight NFP tolerance for cereals and automobiles



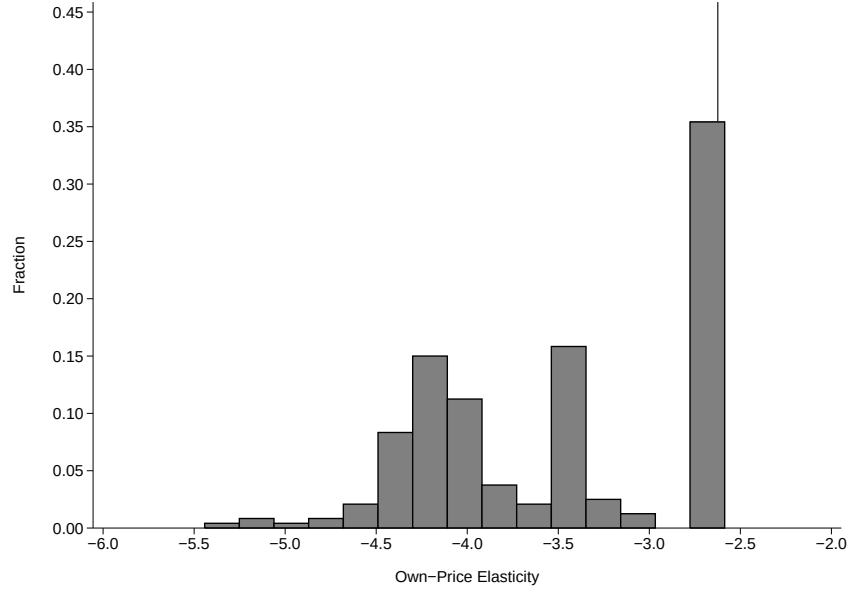
(a) Cereals



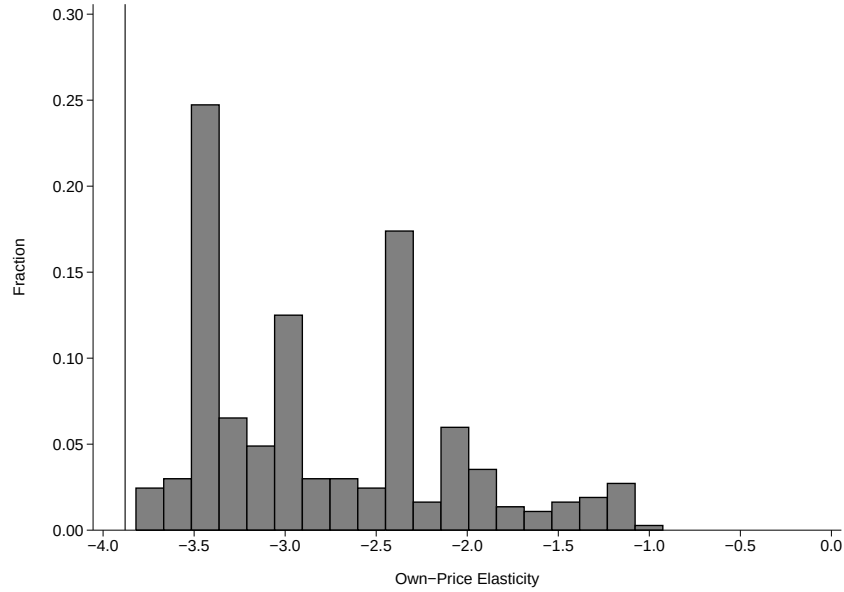
(b) Automobiles

Note: The top product is the product with the largest observed unit market share. For cereals, the top product is brand 5 in market 53 with market share of 45%. For automobiles, the top product is BKRIVE72 in 1972 with market share of 1%. The histograms are based on 240 (383) observations in the case of cereals (automobiles). Observations exceeding 0 are excluded in the case of automobiles. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The vertical lines indicate the value of the own-price elasticity when the objective function value is 4.56 (178.06) for cereal (automobiles).

Figure 3: Own-price elasticity histogram of the median product with tight NFP tolerance for cereals and automobiles



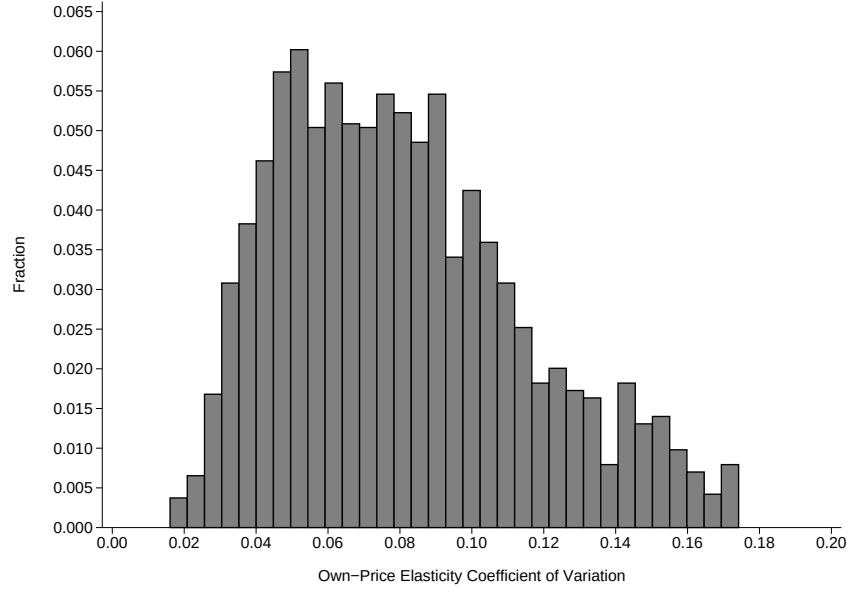
(a) Cereals



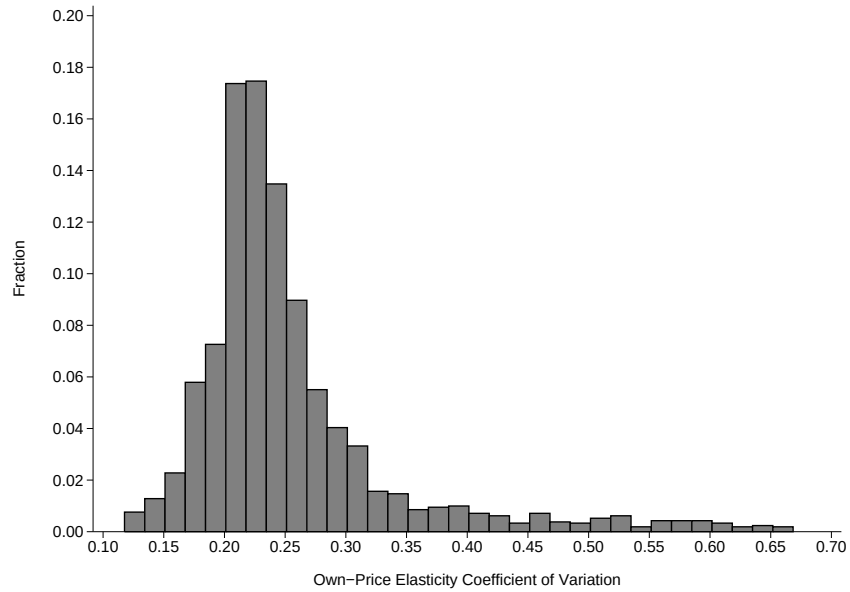
(b) Automobiles

Note: The median product is the product with the median observed unit market share. For cereals, the median product is brand 13 in market 89 with market share of 1%. For automobiles, the median product is NIPULS84 in 1986 with market share of 0.05%. The histograms are based on 240 (368) observations in the case of cereals (automobiles). Observations falling below -3.82 (5th percentile) are excluded in the case of automobiles. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The vertical lines indicate the value of the own-price elasticity when the objective function value is 4.56 (178.06) for cereal (automobiles).

Figure 4: Own-price elasticity coefficient-of-variation histogram of all products with tight NFP tolerance for cereals and automobiles



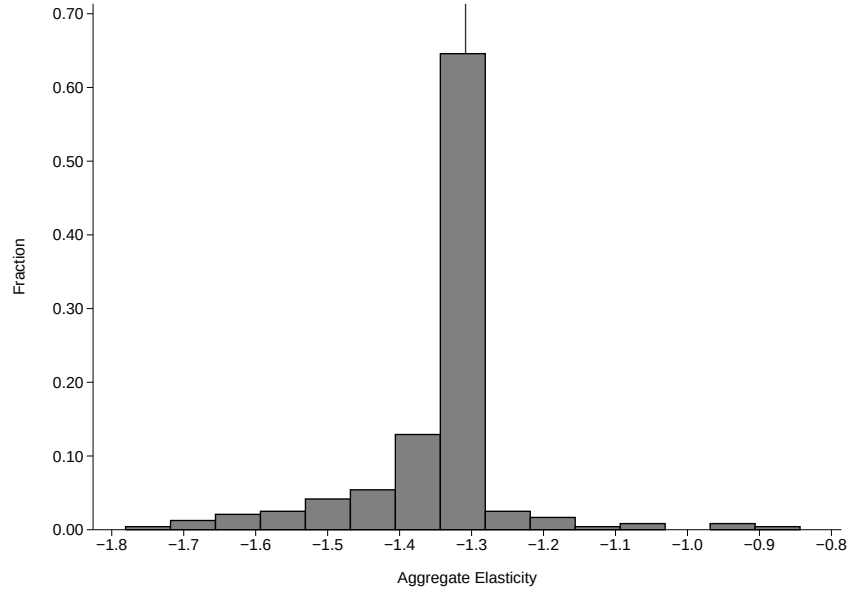
(a) Cereals



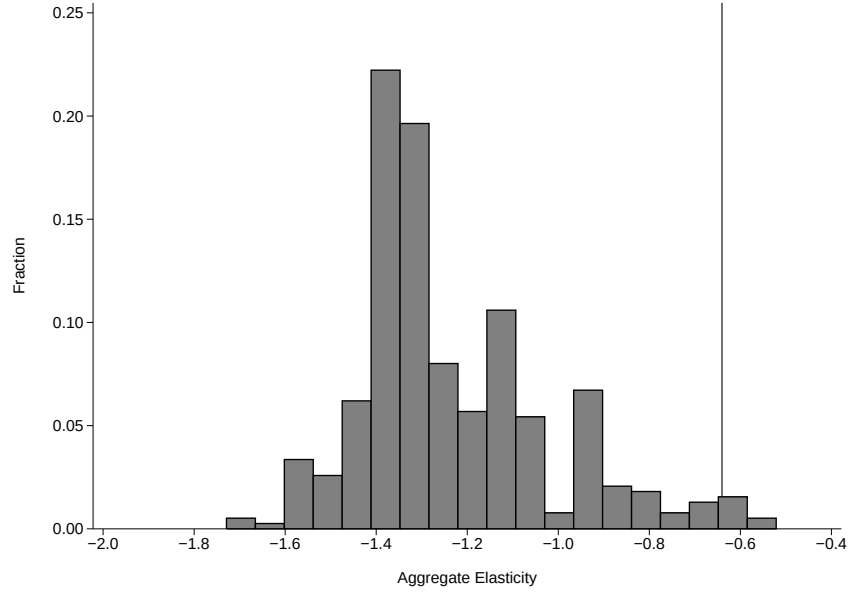
(b) Automobiles

Note: An observation is the ratio of standard deviation to the absolute value of the mean for a product-market combination. The number of observations used to calculate the two moments is 240 (387) for cereals (automobiles). Coefficients of variation exceeding 0.18 (0.67) for cereals (automobiles) are excluded. These upper bounds are the 95th percentiles of the corresponding distributions. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here.

Figure 5: Average aggregate elasticity



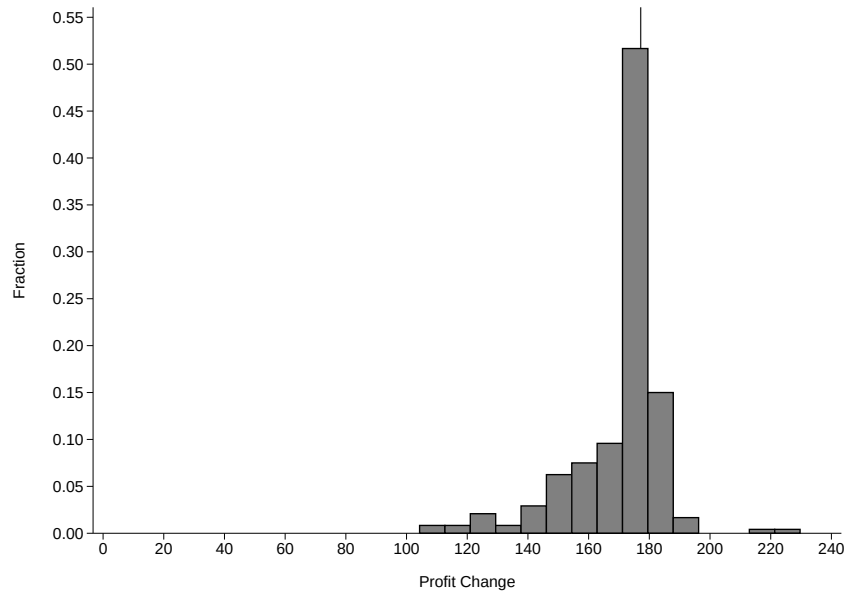
(a) Cereals



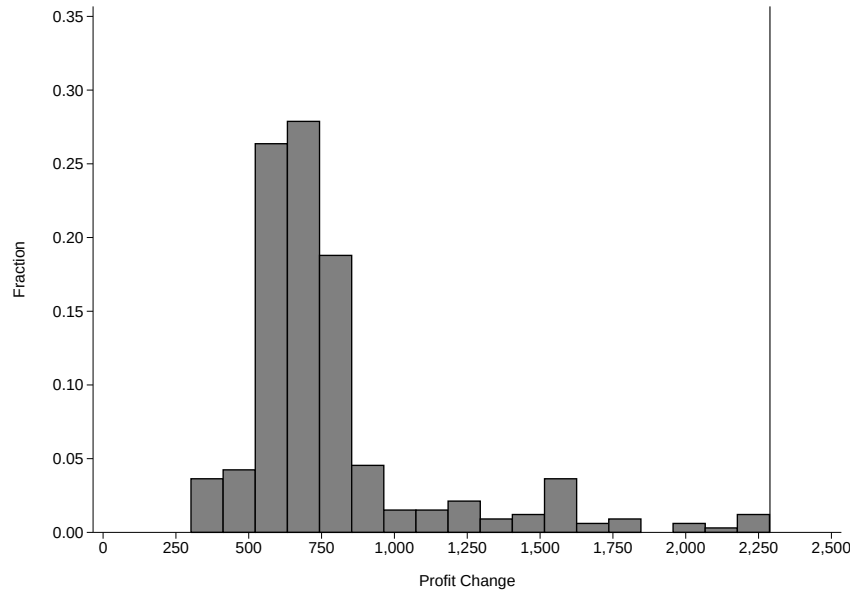
(b) Automobiles

Note: The histogram is based on 240 (387) observations in the case of cereals (automobiles). In the case of automobiles, values above 0 are excluded. Each observation corresponds to a weighted average aggregate elasticity across 94 (19) markets using pre-merger quantities of the inside goods as weights. The automobile market for 1971 is excluded due to extraneous observations. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The details of the aggregate elasticity calculation are available in Section 7.3. The vertical lines indicate the value of the average aggregate elasticity when the objective function value is 4.56 (178.06) for cereal (automobiles).

Figure 6: Average change in profits due to hypothetical mergers (\$millions)



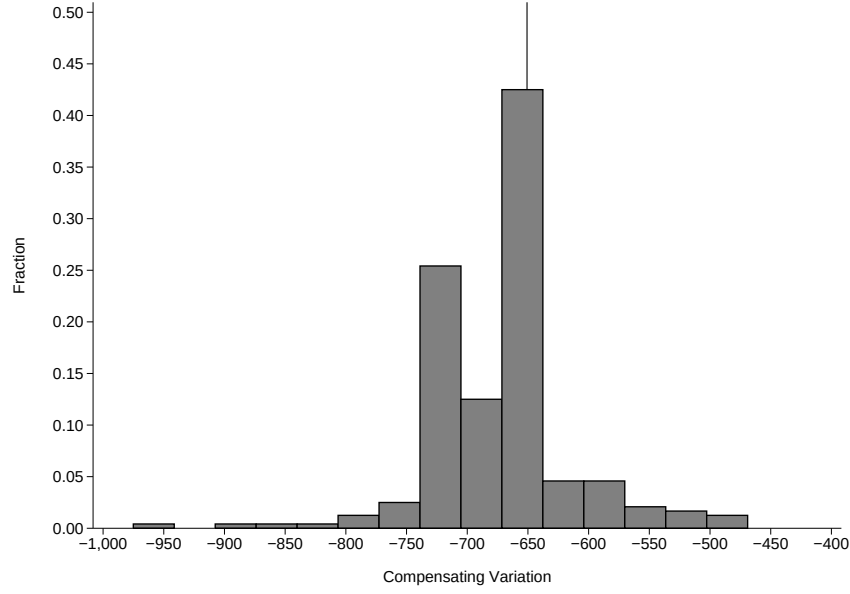
(a) Cereals



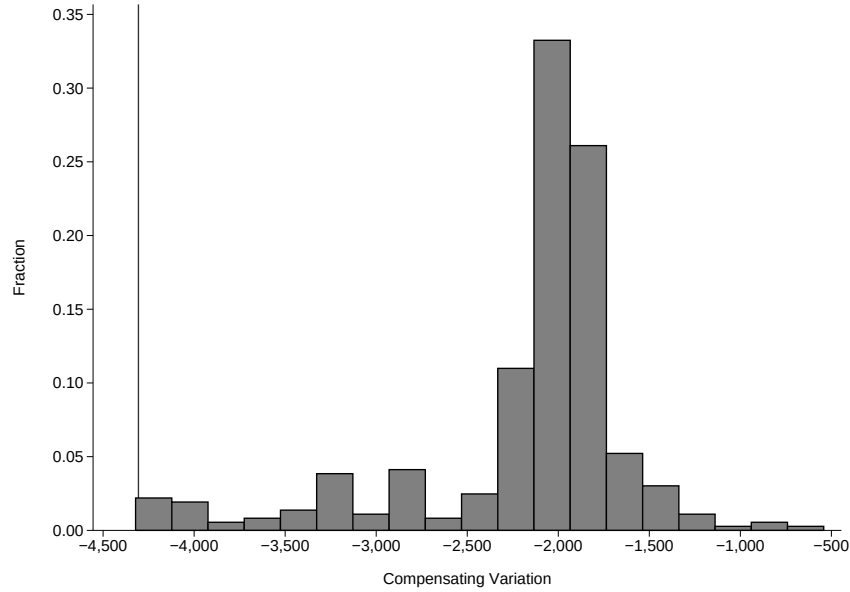
(b) Automobiles

Note: The histogram is based on 240 (329) observations in the case of cereals (automobiles). In the case of automobiles, values below 302m (10th percentile) or above 2,287m (95th percentile) are excluded. Each observation corresponds to a weighted average change in profits across 94 (19) markets using pre-merger quantities of the inside goods as weights. The automobile market for 1971 is excluded due to extraneous observations. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The details of the compensating variation calculation are available in Section 7.3. The vertical lines indicate the value of the average change in profits when the objective function values is 4.56 (178.06) for cereal (automobiles).

Figure 7: Average market compensating variation for hypothetical variation (\$millions)



(a) Cereals



(b) Automobiles

Note: The histogram is based on 240 (364) observations in the case of cereals (automobiles). In the case of automobiles, values below -4,321m (5th percentile) or above 0 are excluded. Each observation corresponds to a weighted average compensating variation across 94 (19) markets using pre-merger quantities of the inside goods as weights. The automobile market for 1971 is excluded due to extraneous observations. The number of observations reflects the sets of starting values that allowed the optimization algorithms to converge under tight NFP tolerance, the removal of extraneous observations discussed in Section 7.1, and any thresholds discussed here. The details of the compensating variation calculation are available in Section 7.3. The vertical lines indicate the value of the average market compensating variation when the objective function values is 4.56 (178.06) for cereal (automobiles).

Table A.1: Parameter estimates by optimization algorithm using loose NFP tolerance for cereals

Objective Function Value	Algorithm	Means			Std. Deviations			Interactions with Demographics									
		Price	Constant	Mushy	Sugar	Price Child	Price Income	Price Income Sq.	Constant Age	Constant Income	Mushy Age	Mushy Income	Sugar Age	Sugar Income			
DER1-QN1	19.552	-30.228 * (6.076)	1.379 (1.511)	0.344 * (0.156)	0.032 (0.256)	-0.008 (0.016)	-0.557 (149.777)	1.017 (4.963)	-0.051 (7.794)	0.920 (1.527)	3.797 * (1.263)	-1.783 (1.787)	1.846 (1.051)	0.030 (0.051)	-0.202 * (0.047)		
DER2-QN2	4.561	-62.739 * (14.802)	3.313 * (1.340)	0.558 * (0.163)	0.093 (0.185)	-0.006 (0.014)	588.481 * (270.414)	11.045 * (4.122)	-30.200 * (14.100)	1.283 * (0.631)	2.294 (1.208)	-1.353 * (0.667)	0.751 (0.802)	0.052 * (0.026)	-0.385 * (0.122)		
DER3-CGR	19.622	-30.911 * (6.006)	1.894 (1.456)	0.386 * (0.161)	-0.034 (0.240)	-0.008 (0.017)	0.168 (145.545)	1.363 (5.015)	-0.032 (7.567)	0.895 (1.511)	3.744 * (1.263)	-1.856 (1.785)	1.994 (1.061)	0.039 (0.050)	-0.212 * (0.046)		
DER4-SOL	4.561	-62.733 * (14.804)	3.313 * (1.340)	0.558 * (0.163)	0.093 (0.185)	-0.006 (0.014)	588.376 * (270.460)	11.055 * (4.123)	-30.195 * (14.102)	1.285 * (0.631)	2.292 (1.209)	-1.353 * (0.667)	0.748 (0.802)	0.052 * (0.026)	-0.385 * (0.122)		
DER5-KNI	4.561	-62.731 * (14.804)	3.312 * (1.340)	0.558 * (0.163)	0.093 (0.185)	-0.006 (0.014)	588.331 * (270.445)	11.054 * (4.123)	-30.192 * (14.101)	1.284 * (0.631)	2.292 (1.209)	-1.353 * (0.667)	0.748 (0.802)	0.052 * (0.026)	-0.385 * (0.122)		
DIR1-SIM	17.224	-30.270 * (8.503)	1.657 (1.177)	0.343 * (0.132)	0.051 (0.197)	-0.003 (0.013)	10.239 (196.155)	4.366 (4.314)	-0.601 (10.194)	0.739 (0.867)	3.838 * (1.085)	-1.283 (1.058)	1.430 (0.754)	0.031 (0.038)	-0.199 * (0.047)		
DIR2-MAD	17.023	-31.286 * (5.905)	1.962 (1.014)	0.356 * (0.136)	-0.011 (0.178)	-0.007 (0.015)	-11.315 (151.628)	9.276 * (4.433)	0.785 (7.944)	0.784 (0.726)	3.283 * (0.943)	-1.091 (0.907)	1.171 (0.646)	0.032 (0.032)	-0.184 * (0.036)		
DIR3-GPS	50.993	-32.821 * (6.009)	1.060 (1.352)	0.407 (0.229)	0.014 (0.447)	-0.096 * (0.028)	2.369 (94.469)	-3.395 (8.713)	-0.008 (4.700)	0.631 (1.859)	3.208 (2.186)	-1.865 (1.953)	2.223 * (1.016)	0.005 (0.067)	-0.213 * (0.052)		
STO1-SIA	131.323	-35.182 * (11.993)	1.848 (0.957)	0.085 (0.147)	-0.886 * (0.321)	-0.011 (0.014)	0.508 (247.209)	0.056 (4.036)	0.507 (12.718)	-0.315 (0.285)	0.916 (1.660)	0.866 (0.973)	0.466 (0.438)	0.131 * (0.054)	-0.023 (0.033)		
STO2-GAL	34.236	-31.041 * (7.344)	2.127 * (1.035)	0.274 * (0.135)	0.325 (0.192)	0.010 (0.012)	0.348 (158.744)	0.377 (4.115)	0.066 (8.179)	0.646 (0.826)	2.437 * (1.091)	-0.950 (0.975)	1.225 (0.652)	0.018 (0.045)	-0.121 * (0.041)		
STO3-SIG	108.129	-34.618 * (2.576)	-2.230 (4.102)	0.527 * (0.218)	0.187 (0.458)	-0.121 (0.088)	-5.323 (132.505)	-1.392 (6.769)	0.771 (7.274)	1.398 (2.250)	2.818 (1.925)	-0.679 (2.716)	0.417 (0.896)	-0.050 (0.086)	-0.203 * (0.063)		
min	4.561	-62.739	1.060	0.274	0.014	0.003	-5.323	-3.395	-30.200	0.631	2.292	-1.865	0.417	-0.050	-0.385		
max	108.129	-30.228	3.313	0.558	0.325	0.121	588.481	11.055	0.771	1.398	3.838	-0.679	2.223	0.052	-0.121		

Note: We report results for the set of starting values that give rise to the minimum objective function value for each algorithm. The naming convention for the optimization algorithms in the “Algorithm” column follows Table 2. We implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. DIR2-MAD and STO1-SIA produced exit codes consistent with exceeding the maximum number of function evaluations. The column “Means” contains coefficient estimates obtained using linear IVs. The columns “Std. Deviations” and “Interactions with Demographics” contain coefficient estimates obtained using the nonlinear search in the outer loop of the BLP algorithm. Std. errors are reported in parentheses. The asterisk (*) indicates statistical significance at 5% level. The minimum and maximum for the coefficients in the column Std. Deviations are calculated using absolute values and exclude the entries associated with DIR2-MAD and STO1-SIA.

Table A.2: Parameter estimates by optimization algorithm using tight NFP tolerance for cereals

Algorithm	Objective Function Value	Means			Std. Deviations			Interactions with Demographics											
		Price		Child	Price		Income Sq.	Constant		Mushy Age	Mushy Income		Sugar Age	Sugar Income					
		19.554	4.562		19.929	4.562	4.562	19.929	4.562	19.929	4.562	19.929	4.562	19.929	4.562	19.929	4.562	19.929	4.562
DER1-QN1	19.554	-30.251 *	1.369	0.345 *	0.030	-0.008	1.015	-0.557	-0.047	0.917	3.788 *	-1.779	1.836	0.031	-0.202 *				
		(6.070)	(1.507)	(0.156)	(0.254)	(0.016)	(4.953)	(149.689)	(7.790)	(1.516)	(1.255)	(1.777)	(1.047)	(0.050)	(0.047)				
DER2-QN2	4.562	-62.732 *	3.312 *	0.558 *	0.093	-0.006	11.061 *	588.415 *	-30.197 *	1.286 *	2.293	-1.355 *	0.748	0.052 *	-0.385 *				
		(14.806)	(1.341)	(0.163)	(0.185)	(0.014)	(4.123)	(270.504)	(14.105)	(0.632)	(1.209)	(0.667)	(0.802)	(0.026)	(0.122)				
DER3-CGR	19.929	-30.215 *	1.510	0.331 *	0.086	-0.009	0.868	-0.142	-0.091	1.150	3.732 *	-2.072	1.972	0.027	-0.201 *				
		(5.352)	(1.508)	(0.164)	(0.289)	(0.016)	(5.340)	(126.945)	(6.591)	(1.781)	(1.474)	(1.994)	(1.055)	(0.054)	(0.049)				
DER4-SOL	4.562	-62.730 *	3.313 *	0.558 *	0.093	-0.006	11.055 *	588.327 *	-30.192 *	1.285 *	2.292	-1.353 *	0.748	0.052 *	-0.385 *				
		(14.803)	(1.340)	(0.163)	(0.185)	(0.014)	(4.123)	(270.442)	(14.101)	(0.631)	(1.209)	(0.667)	(0.802)	(0.026)	(0.122)				
DER5-KNI	4.562	-62.737 *	3.313 *	0.558 *	0.094	-0.006	11.055 *	588.463 *	-30.199 *	1.285 *	2.292	-1.354 *	0.748	0.052 *	-0.385 *				
		(14.806)	(1.340)	(0.163)	(0.185)	(0.014)	(4.123)	(270.488)	(14.104)	(0.631)	(1.209)	(0.667)	(0.802)	(0.026)	(0.122)				
DIR1-SIM	17.549	-29.398 *	1.296	0.338 *	0.189	-0.004	6.725	2.702	-0.251	0.963	3.906 *	-1.503	1.662	0.029	-0.195 *				
		(7.681)	(1.314)	(0.135)	(0.238)	(0.014)	(4.915)	(187.724)	(9.794)	(1.082)	(1.190)	(1.251)	(0.857)	(0.042)	(0.047)				
DIR2-MAD	16.055	-31.610 *	1.968	0.345 *	0.021	-0.003	7.507	5.105	-0.125	0.797	3.336 *	-1.208	1.354 *	0.031	-0.191 *				
		(7.789)	(1.034)	(0.127)	(0.185)	(0.012)	(4.375)	(176.345)	(9.162)	(0.793)	(1.077)	(0.947)	(0.656)	(0.035)	(0.043)				
DIR3-GPS	50.993	-32.821 *	1.060	0.407	0.014	-0.096 *	-3.395	2.369	-0.008	0.631	3.208	-1.865	2.223 *	0.005	-0.213 *				
		(6.009)	(1.352)	(0.229)	(0.447)	(0.028)	(8.713)	(94.469)	(4.700)	(1.859)	(2.186)	(1.953)	(1.016)	(0.067)	(0.052)				
STO1-SIA	131.320	-35.182 *	1.848	0.085	-0.886 *	-0.011	0.056	0.508	0.507	-0.315	0.916	0.866	0.466	0.131 *	-0.023				
		(11.993)	(0.957)	(0.147)	(0.321)	(0.014)	(4.036)	(247.210)	(12.718)	(0.285)	(1.660)	(0.973)	(0.438)	(0.054)	(0.033)				
STO2-GAL	31.594	-31.995 *	0.732	0.060	-0.036	-0.009	2.697	-0.698	0.143	0.786	3.650 *	-2.189	2.204 *	0.077 *	-0.275 *				
		(2.219)	(1.121)	(0.156)	(0.209)	(0.018)	(5.370)	(45.260)	(2.400)	(0.884)	(0.892)	(1.246)	(0.850)	(0.030)	(0.035)				
STO3-SIG	108.130	-34.618 *	-2.230	0.527 *	0.187	-0.121	-1.392	-5.323	0.771	1.398	2.818	-0.679	0.417	-0.050	-0.203 *				
		(2.576)	(4.102)	(0.218)	(0.458)	(0.088)	(6.769)	(132.505)	(7.274)	(2.250)	(1.925)	(2.716)	(0.896)	(0.086)	(0.063)				
min	4.562	-62.737	0.732	0.060	0.014	0.004	-3.395	-5.323	-30.199	0.631	2.292	-2.189	0.417	-0.050	-0.385				
max	108.130	-29.398	3.313	0.558	0.189	0.121	11.061	588.463	0.771	1.398	3.906	-0.679	2.223	0.077	-0.195				

Note: We report results for the set of starting values that give rise to the minimum objective function value for each algorithm. The naming convention for the optimization algorithms in the “Algorithm” column follows Table 2. We implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. DIR2-MAD and STO1-SIA produced exit codes consistent with exceeding the maximum number of function evaluations. DER5-KNI produced the exit code -100. According to Appendix A in [Waltz and Plantenga \(2009\)](#), the exit code -100 indicates that “the stopping tests are close to being satisfied (within a factor of 100) and so the approximate solution is believed to be optimal.” The column “Means” contains coefficient estimates obtained using linear IVs. The columns “Std. Deviations” and “Interactions with Demographics” contain coefficient estimates obtained using the nonlinear search in the outer loop of the BLP algorithm. Std. errors are reported in parentheses. The asterisk (*) indicates statistical significance at 5% level. The minimum and maximum for the coefficients in the column Std. Deviations are calculated using absolute values and exclude the entries associated with DIR2-MAD and STO1-SIA.

Table A.3: Parameter estimates by optimization algorithm using loose NFP tolerance for automobiles

Objective Function		Means					Std. Deviations					
Algorithm	Value	Price	Constant	HP/WT	AC	MPG	Space	Price	Constant	HP/WT	AC	MPG
DER1-QN1	214.999	-0.458 * (0.189)	-9.751 * (1.153)	3.064 * (0.841)	1.266 * (1.123)	-0.611 (0.762)	2.762 * (0.287)	0.183 (0.109)	-2.718 * (0.816)	1.015 (1.212)	0.541 (5.005)	0.686 (0.447)
DER2-QN2	198.203	-0.554 * (0.121)	-14.693 * (1.527)	2.129 (2.338)	-5.151 (3.645)	0.023 (0.191)	3.049 * (0.350)	-0.293 * (0.091)	6.522 * (1.783)	-2.886 (2.675)	-7.823 * (3.217)	0.048 (0.392)
DER3-CGR	215.056	-0.464 * (0.206)	-9.671 * (1.228)	3.052 * (0.963)	1.288 (0.955)	-0.635 (0.844)	2.763 * (0.308)	0.185 (0.116)	-2.710 * (0.819)	1.039 (1.254)	0.476 (5.409)	0.699 (0.495)
DER4-SOL	214.969	-0.456 * (0.200)	-9.751 * (1.223)	3.037 * (0.947)	1.277 (1.002)	-0.613 (0.815)	2.759 * (0.297)	0.182 (0.113)	-2.721 * (0.832)	1.063 (1.241)	0.485 (5.399)	0.687 (0.481)
DER5-KNI	277.621	-0.361 * (0.066)	-6.879 * (0.872)	1.991 * (0.758)	0.883 * (0.291)	-2.444 * (0.792)	2.616 * (0.200)	0.141 * (0.035)	0.000 (0.766)	0.125 (2.152)	0.552 (0.687)	1.668 * (0.428)
DIR1-SIM	215.040	-0.458 * (0.188)	-9.728 * (1.123)	3.016 * (0.828)	1.266 (1.216)	-0.642 (0.724)	2.749 * (0.272)	0.183 (0.110)	-2.767 * (0.858)	1.164 (1.185)	0.560 (5.195)	0.703 (0.421)
DIR2-MAD	215.053	-0.451 * (0.215)	-9.763 * (1.264)	3.067 * (0.970)	1.279 (0.960)	-0.623 (0.890)	2.756 * (0.321)	0.179 (0.121)	-2.709 * (0.824)	0.993 (1.318)	0.453 (5.741)	0.693 (0.522)
DIR3-GPS	215.064	-0.451 * (0.195)	-9.775 * (1.278)	2.985 * (1.089)	1.284 (0.741)	-0.597 (0.828)	2.757 * (0.299)	0.179 (0.109)	-2.709 * (0.811)	1.134 (1.345)	0.412 (5.469)	0.677 (0.495)
STO1-SIA	215.953	-0.430 * (0.160)	-9.920 * (1.266)	2.985 * (1.065)	1.258 * (0.393)	-0.487 (0.787)	2.772 * (0.299)	0.168 (0.087)	-2.541 * (0.671)	0.925 (1.566)	0.327 (4.396)	0.617 (0.481)
STO2-GAL	215.505	-0.438 * (0.176)	-9.939 * (1.257)	3.071 * (0.979)	1.263 (0.654)	-0.514 (0.818)	2.774 * (0.312)	0.173 (0.097)	-2.632 * (0.706)	0.848 (1.516)	0.403 (4.678)	0.631 (0.496)
STO3-SIG	210.466	-0.397 * (0.058)	-6.018 * (1.031)	-1.312 (1.071)	0.420 (0.391)	-4.057 * (0.970)	1.884 * (0.353)	0.151 * (0.028)	4.294 * (0.964)	-6.241 * (0.837)	2.030 * (0.483)	2.652 * (0.544)
min	198.203	-0.554	-14.693	-1.312	-5.151	-4.057	1.884	0.141	0.000	0.125	0.403	0.048
max	277.621	-0.361	-6.018	3.071	1.288	0.023	3.049	0.293	6.522	6.241	7.823	2.652

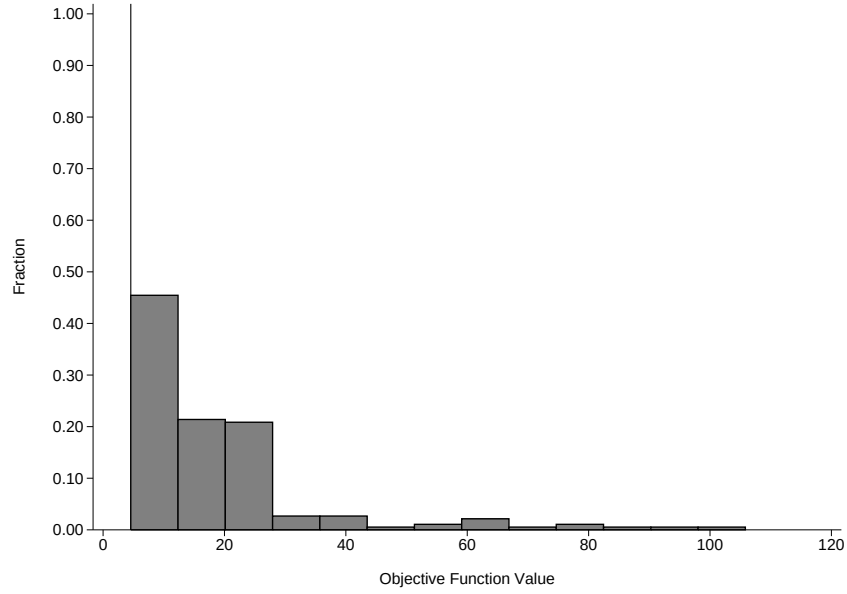
Note: We report results for the set of starting values that give rise to the minimum objective function value for each algorithm. The naming convention for the optimization algorithms in the “Algorithm” column follows Table 2. We implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. DIR1-SIM and STO1-SIA produced exit codes consistent with exceeding the maximum number of function evaluations. DER5-KNI produced the exit code -100. According to Appendix A in [Waltz and Plantenga \(2009\)](#), the exit code -100 indicates that “the stopping tests are close to being satisfied (within a factor of 100) and so the approximate solution is believed to be optimal.” The column “Means” contains coefficient estimates obtained using linear IVs. The columns “Std. Deviations” contain coefficient estimates obtained using the nonlinear search in the outer loop of the BLP algorithm. Std. errors are reported in parentheses. The asterisk (*) indicates statistical significance at 5% level. The minimum and maximum for the coefficients in the column Std. Deviations are calculated using absolute values and exclude the entries associated with DIR1-SIM and STO1-SIA.

Table A.4: Parameter estimates by optimization algorithm using tight NFP tolerance for automobiles

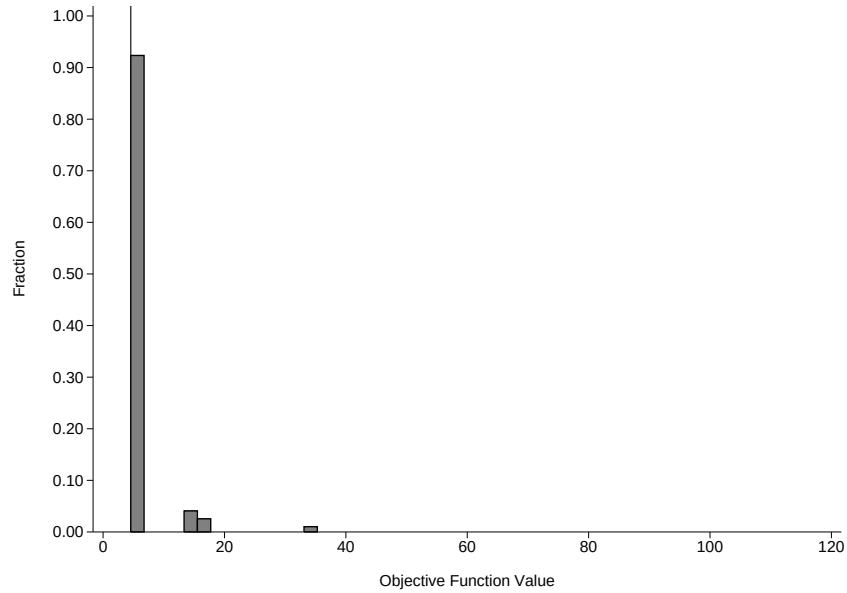
Algorithm	Objective Function Value	Means						Std. Deviations					
		Price			HP/WT			Constant			HP/WT		
		Price	Constant	MPG	AC	MPG	Space	Price	Constant	MPG	AC	MPG	Space
DER1-QN1	215.091	-0.456 *	-9.753 *	3.038 *	1.277	-0.613	2.760 *	0.182	-2.719 *	1.063	0.484	0.686	
		(0.199)	(1.225)	(0.948)	(0.996)	(0.815)	(0.297)	(0.113)	(0.830)	(1.243)	(5.389)	(0.481)	
DER2-QN2	178.061	-0.409 *	-5.668 *	-2.601 *	0.692	-4.540 *	1.738 *	0.156 *	5.178 *	-7.192 *	1.645 *	2.913 *	
		(0.064)	(1.170)	(1.150)	(0.355)	(0.984)	(0.375)	(0.031)	(0.908)	(0.862)	(0.444)	(0.538)	
DER3-CGR	215.163	-0.454 *	-9.714 *	3.021 *	1.276	-0.609	2.760 *	0.180	-2.659 *	1.041	0.463	0.685	
		(0.200)	(1.251)	(0.971)	(0.919)	(0.842)	(0.303)	(0.113)	(0.820)	(1.263)	(5.364)	(0.495)	
DER4-SOL	178.060	-0.409 *	-5.670 *	-2.608 *	0.691	-4.541 *	1.738 *	0.156 *	5.181 *	-7.194 *	1.646 *	2.914 *	
		(0.064)	(1.170)	(1.151)	(0.355)	(0.984)	(0.375)	(0.031)	(0.908)	(0.862)	(0.444)	(0.538)	
DER5-KNI	277.733	-0.360 *	-6.915 *	2.010 *	0.888 *	-2.408 *	2.615 *	0.141 *	0.000	0.089	0.546	1.649 *	
		(0.067)	(0.874)	(0.740)	(0.290)	(0.794)	(0.201)	(0.035)	(0.773)	(2.145)	(0.686)	(0.429)	
DIR1-SIM	178.060	-0.409 *	-5.671 *	-2.608 *	0.690	-4.540 *	1.738 *	0.156 *	5.180 *	-7.193 *	1.646 *	2.913 *	
		(0.064)	(1.170)	(1.150)	(0.355)	(0.984)	(0.375)	(0.031)	(0.907)	(0.862)	(0.444)	(0.538)	
DIR2-MAD	208.064	-0.583 *	-15.386 *	2.883	-5.291	-0.016	3.083 *	-0.305 *	7.024 *	-2.394	-8.118 *	-0.096	
		(0.124)	(1.602)	(1.694)	(3.435)	(0.203)	(0.377)	(0.089)	(1.779)	(2.036)	(3.076)	(0.281)	
DIR3-GPS	178.060	-0.409 *	-5.672 *	-2.607 *	0.690	-4.538 *	1.738 *	0.156 *	5.179 *	-7.192 *	1.646 *	2.912 *	
		(0.064)	(1.170)	(1.150)	(0.355)	(0.984)	(0.375)	(0.031)	(0.907)	(0.862)	(0.444)	(0.538)	
STO1-SIA	191.194	-0.353 *	-5.832 *	-2.411 *	0.533	-4.457 *	1.826 *	0.127 *	4.593 *	-6.564 *	1.683 *	2.865 *	
		(0.059)	(1.056)	(1.166)	(0.357)	(0.894)	(0.345)	(0.030)	(0.876)	(0.825)	(0.434)	(0.491)	
STO2-GAL	205.434	-0.284 *	-7.387 *	-1.760 *	0.555	-2.939 *	2.081 *	0.089 *	3.642 *	-5.318 *	1.346 *	2.075 *	
		(0.050)	(1.035)	(0.893)	(0.307)	(0.848)	(0.287)	(0.028)	(0.741)	(0.689)	(0.394)	(0.459)	
STO3-SIG	186.234	-0.384 *	-6.306 *	-1.992	0.440	-3.920 *	1.874 *	0.144 *	4.640 *	-6.438 *	1.888 *	2.576 *	
		(0.059)	(1.139)	(1.051)	(0.373)	(0.987)	(0.353)	(0.029)	(0.870)	(0.798)	(0.417)	(0.538)	
min	178.060	-0.456	-9.753	-2.608	0.440	-4.541	1.738	0.089	0.000	0.089	0.463	0.685	
max	277.733	-0.284	-5.668	3.038	1.277	-0.609	2.760	0.182	5.181	7.194	1.888	2.914	

Note: We report results for the set of starting values that give rise to the minimum objective function value for each algorithm. The naming convention for the optimization algorithms in the “Algorithm” column follows Table 2. We implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. DIR2-MAD and STO1-SIA produced exit codes consistent with exceeding the maximum number of function evaluations. The column “Means” contains coefficient estimates obtained using linear IVs. The columns “Std. Deviations” contain coefficient estimates obtained using the nonlinear search in the outer loop of the BLP algorithm. Std. errors are reported in parentheses. The asterisk (*) indicates statistical significance at 5% level. The minimum and maximum for the coefficients in the column Std. Deviations are calculated using absolute values and exclude the entries associated with DIR2-MAD and STO1-SIA.

Figure A.1: Histogram of the objective function value for derivative-based algorithms under tight NFP tolerance for cereals



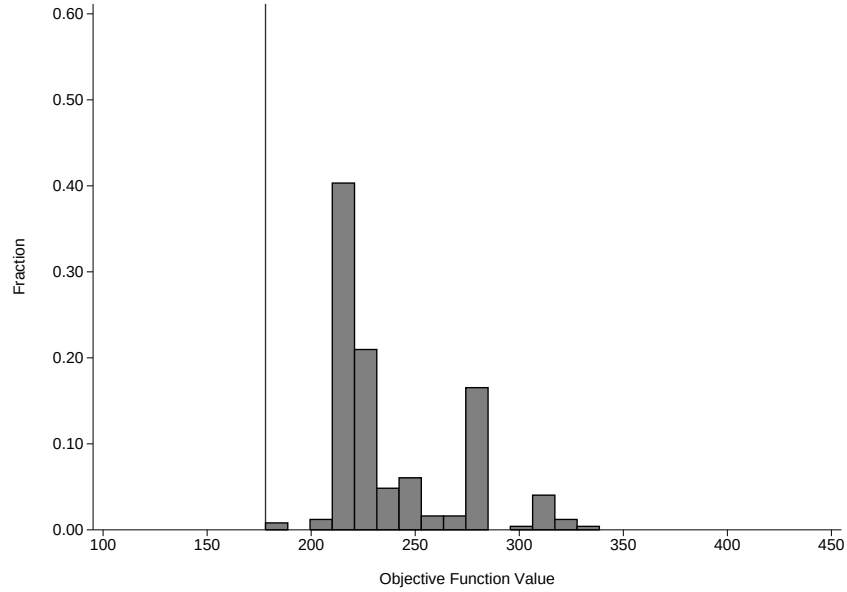
(a) Outer tolerance 1E-03



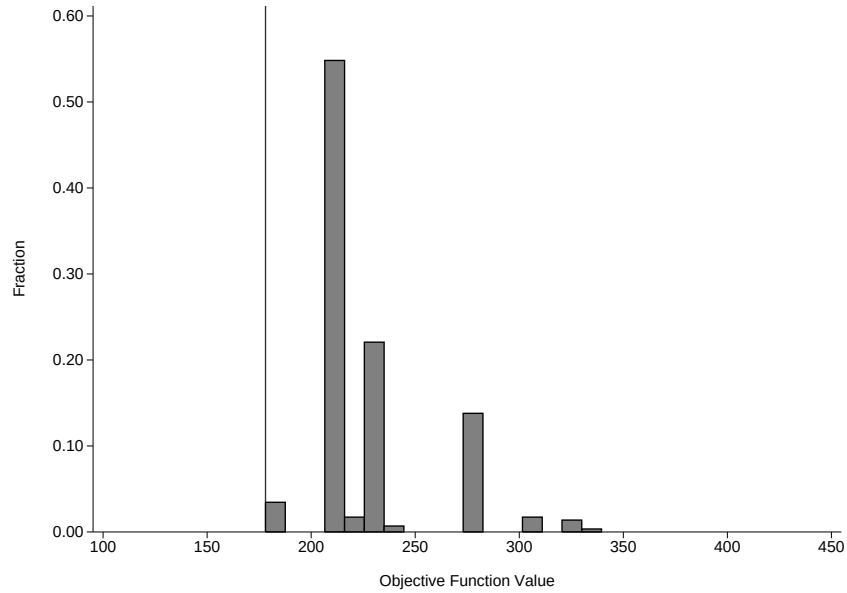
(b) Outer tolerance 1E-06

Note: The vertical solid line indicates the minimum objective function value for the combinations of optimization algorithms and starting values that converged. In panel (a), we implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. In panel (b), we implemented DER5-KNI with and without a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. Similar to Figure 1, both histograms exclude objective function values exceeding 110.

Figure A.2: Histogram of the objective function value for derivative-based algorithms under tight NFP tolerance for automobiles



(a) Outer tolerance 1E-03



(b) Outer tolerance 1E-06

Note: The vertical solid line indicates the minimum objective function value for the combinations of optimization algorithms and starting values that converged. In panel (a), we implemented DER5-KNI with a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. In panel (b), we implemented DER5-KNI with and without a lower bound of zero on the parameter vector over which the algorithm performed the nonlinear search. Similar to Figure 1, both histograms exclude objective function values exceeding 430.